Todays Session Details

Challenges & Opportunities: Analog VLSI Design

Activity Based Learning

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Professor, Dept. of ECE, Cambridge Institute of Technology, KR Puram, Bangalore

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ENTREPRENEUR

Career Opportunities

- Job Posting – VLSI Circuit Design Engineer (Multiple Openings)
- LOCATIONS: Burlington, VT / Tempe, AZ / Raleigh, NC / *Fishkill, NY / Austin, TX, Boxborough, MA / Fort Collins, CO*
- Experience with designing several of the following circuit families, including SERDES, DDR, VREG, PLL, DLL, ADC/DAC, Standard Cells, and etc.
- Familiarity with CAD tools/flows for circuit analysis (e.g., IR drop, EM, ERC checker, Timing analysis, etc.)
- Scripting languages like Perl/Python



Chips to Startup Programme

CAMBRIDGE INSTITUTE OF TECHNOLOGY

CCCIR CAMBRIAN

SYNOPSYS

Texas Instruments India

Analog IC Design Engineer

We are hiring one lead level IC Design Engineer with demonstrated design cycle experience. Some of the possible roles are:

- PLL & Clock System Design
- IF filter and VGA Design
- Transmitter Design

Candidate should have gone through 1 complete design cycle of an RF transceiver: namely Architecture, Schematic Design, Layout Design, Tape Out, Test & Validation and identifying design fixes for subsequent Tape Out.

Analog design

Analog Layout Engineer

Accenture · 3.9 ★

Bengaluru, Karnataka

Analog Circuit Design

Mirafra Software Technologies ·

India

Analog Layout Design Engineer

Pozibility Technologies · 3.7 ★

Marvell Technology

Analog Layout Staff Engineer

Education: BE/B.Tech or MS/M.Tech in Electrical/Electronics Engineering, Microelectronics, or related fields; 3-4 years of experience in high-speed analog/custom layout development.
Strong understanding of semiconductor process technologies, device physics, and layout effects in advanced nodes.
Proven experience in full-custom circuit layout and verification, including RC extraction, physical verification (DRC, LVS, ERC, DFM), and EMIR analysis.
Hands-on expertise in mixed-signal/analog/high-speed layouts (SerDes, ADC/DAC, PLL, etc.) across advanced FinFET technologies.
Proficiency with Cadence Virtuoso and industry-standard EDA tools; scripting skills (Perl, Tcl, SKILL) for automation are a plus.

Staff Analog Design Engineer

Renesas Electronics India Pvt. Ltd.

Analog Layout Engineer

Digicomm Semiconductor · 2.7 ★

Bengaluru, Karnataka

Principal Engineer, Analog Design

Globalfoundries ★ 3.4 | 130 Reviews

Analog Design Engineer

Onsemi Vietnam ★ 3.3 | 38 I

Analog Design Engineer (All Levels)

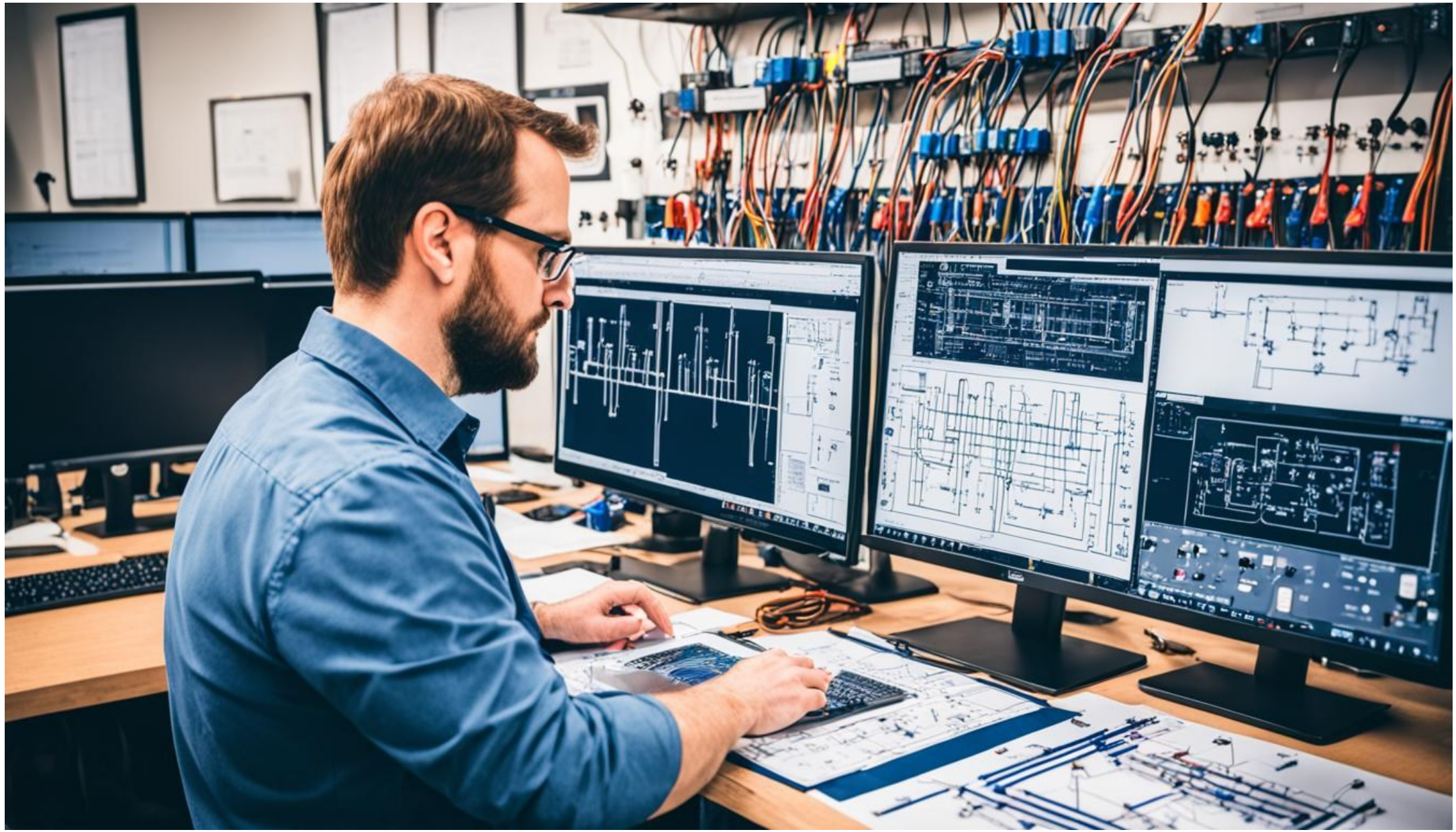
Blueberry Semiconductors ★ 2.1 | 25 Reviews

Chips to Startup Programme

CAMBRIDGE INSTITUTE OF TECHNOLOGY

CAMBRIAN

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VLSI curriculum

1992 – CMOS circuits by KANG or by PUCKNELL or by NEIL H E WESTE / DAVID HARRIS / AYAN BANERJEE (Edition)

2024 – CMOS circuits by KANG or PUCKNELL or by NEIL H E WESTE / DAVID HARRIS / AYAN BANERJEE (Edition)

1996

SAR ADC using Cadence

ALU Design on FPGA
Xilinx/ Altera
Spartan-3/ Virtex

2024

SAR ADC using Cadence

ALU Design on FPGA
Xilinx/ Altera
Spartan-3/ Virtex

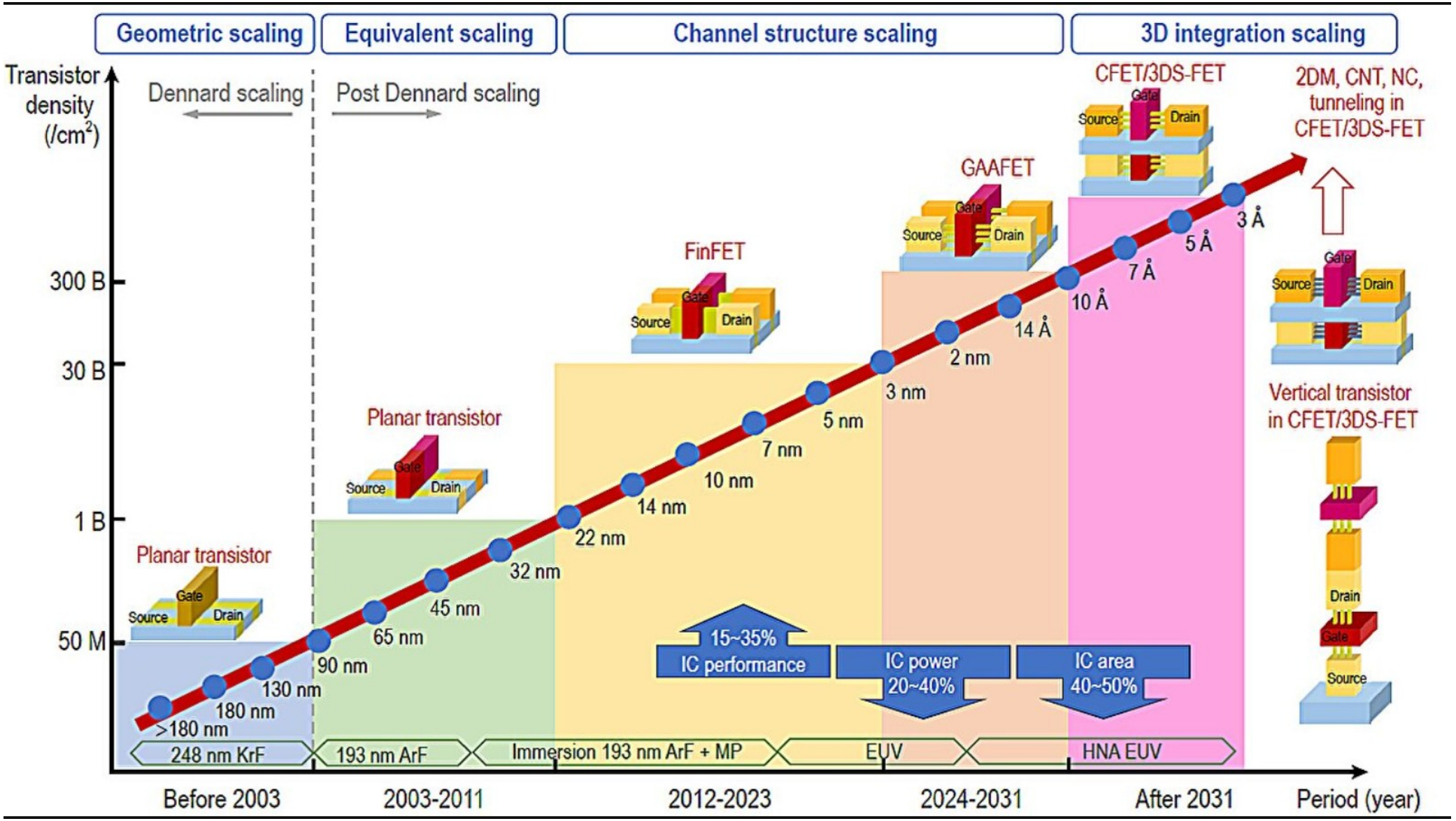
2026

6T RAM
UART using Cadence

ALU Design on FPGA
Xilinx/ Altera
Spartan-3/ Virtex

202x

Transistor technology roadmap



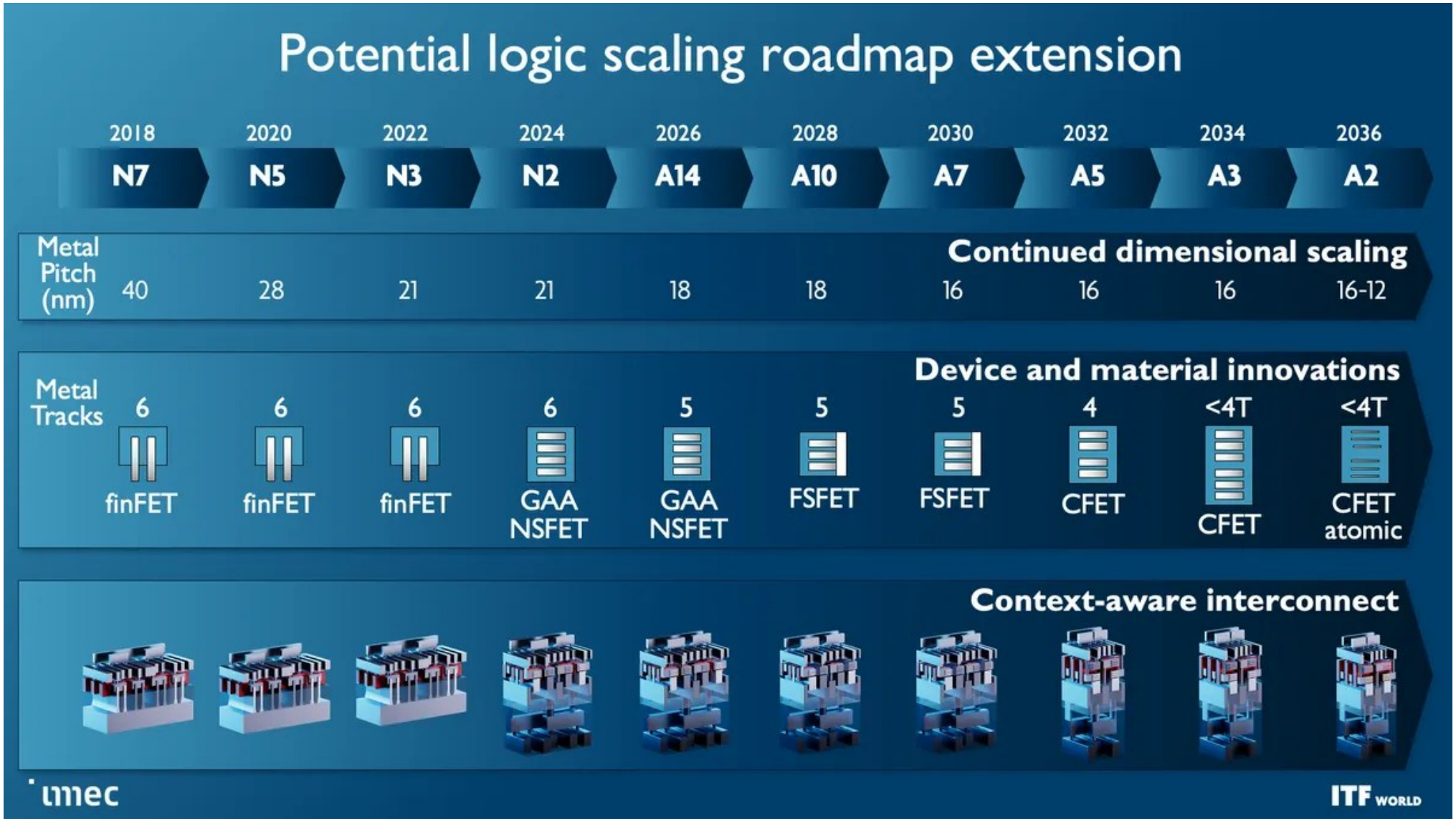
Chips to Startup Programme

CAMBRIDGE INSTITUTE OF TECHNOLOGY

CAMBRIAN

SYNOPSYS

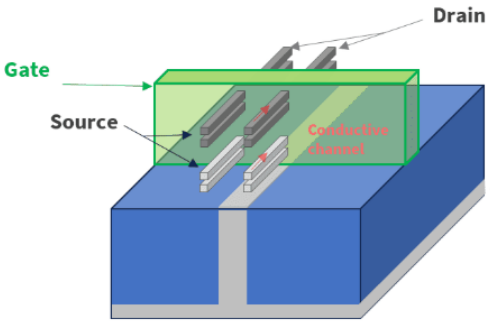
Transistor Technology



Floating Source Field-Effect Transistor

Nanosheet Field-Effect Transistor

Complementary Field-Effect Transistor



Chips to Startup Programme

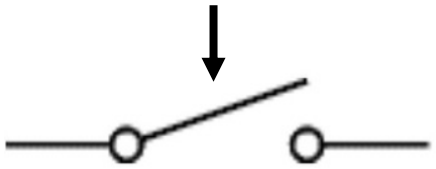
CAMBRIDGE INSTITUTE OF TECHNOLOGY

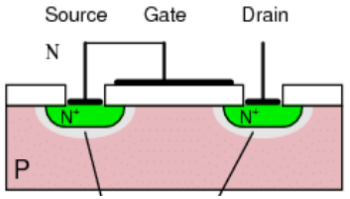
CCCIR CAMBRIAN

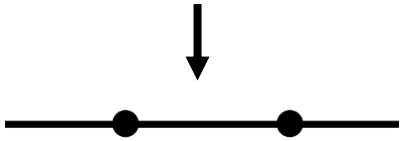
SYNOPSYS®

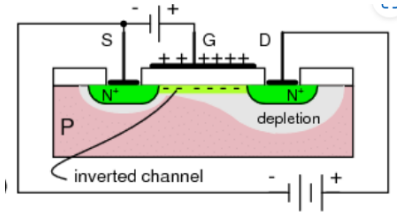
VLSI basics

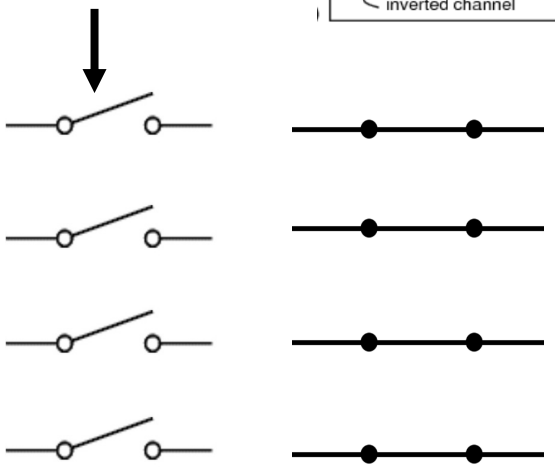
- Transistor is a switch

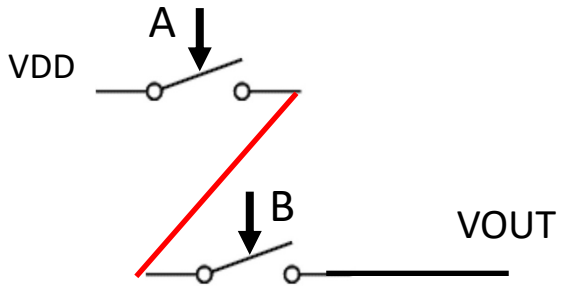




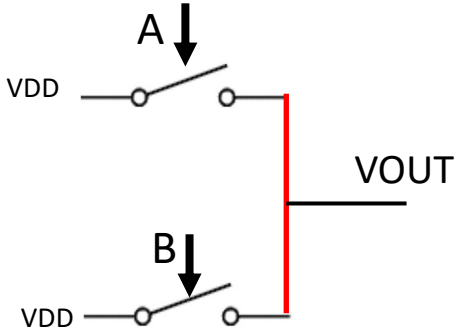






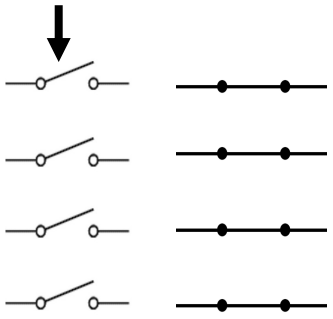


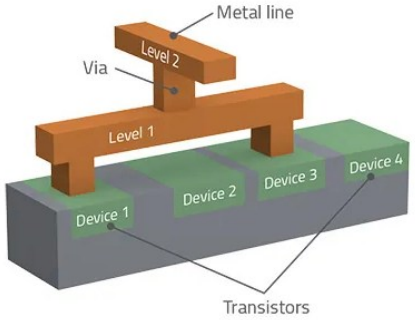
A	B	VOUT
0	0	0
0	1	0
1	0	0
1	1	1

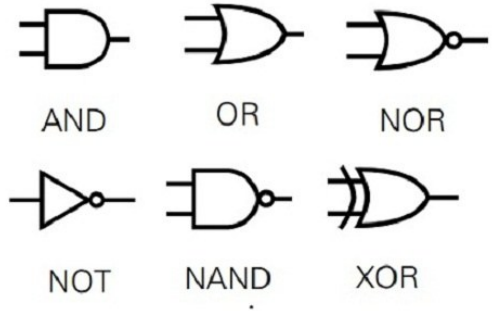


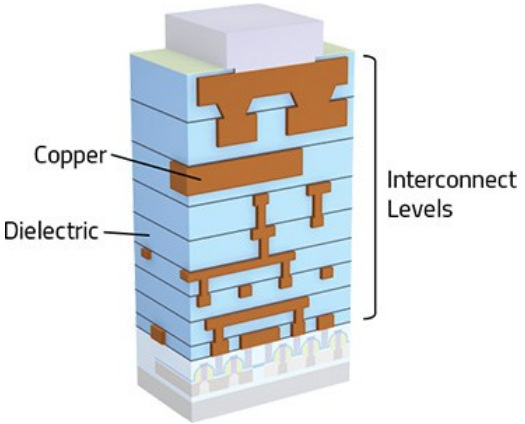
A	B	VOUT
0	0	0
0	1	1
1	0	1
1	1	1

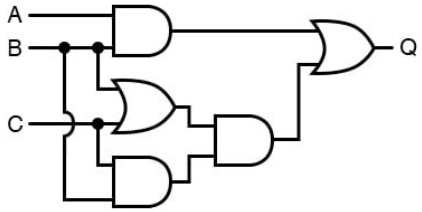
VLSI basics





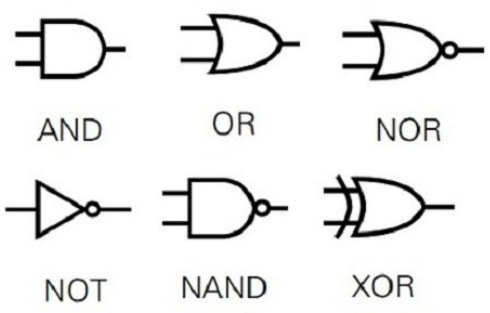


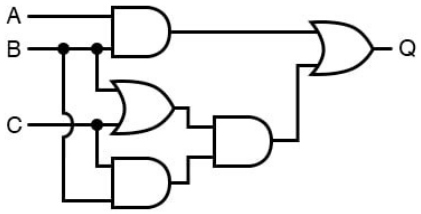


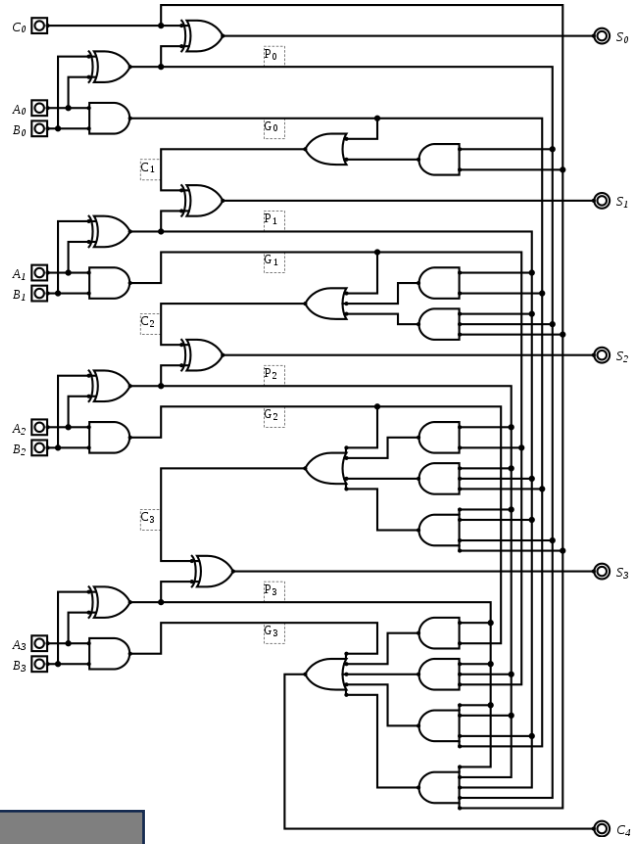


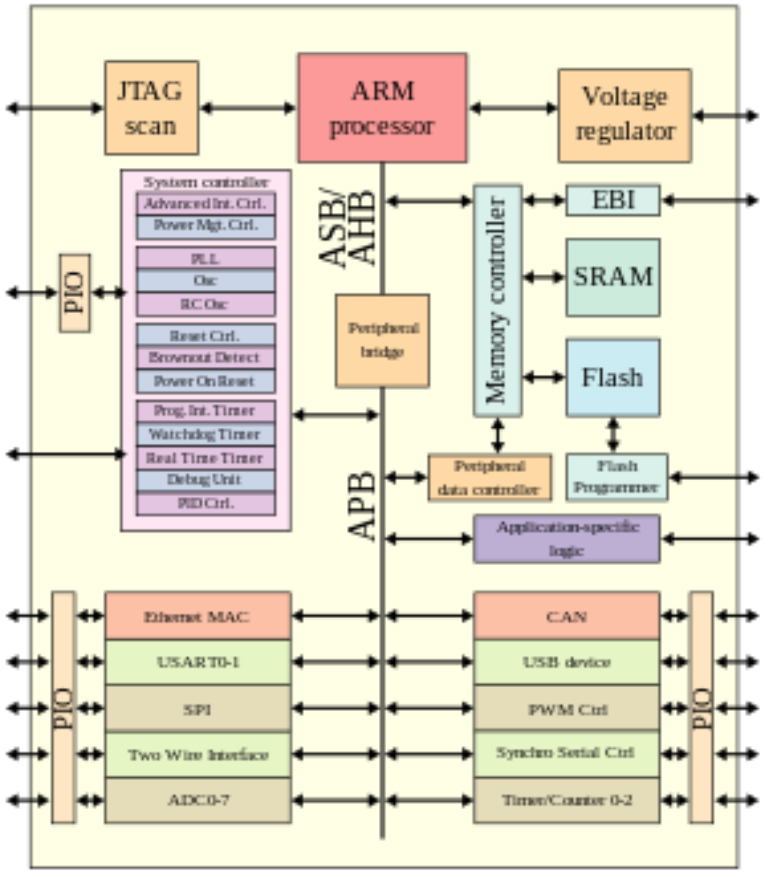
Transistors + Interconnects Gates + Interconnects Logic

VLSI Circuits



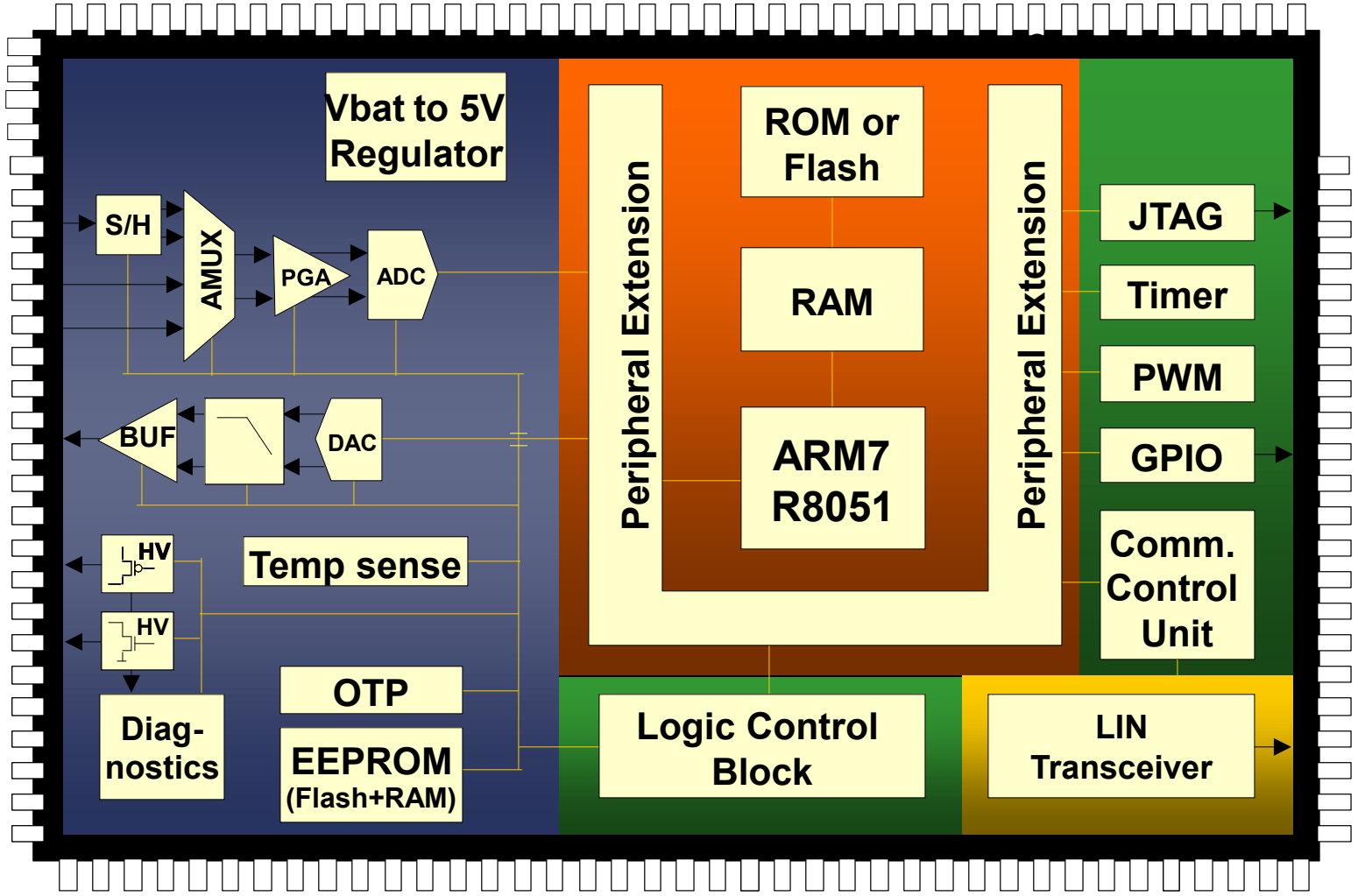






Logic + Interconnects → Subsystems + Interconnects → Complex System

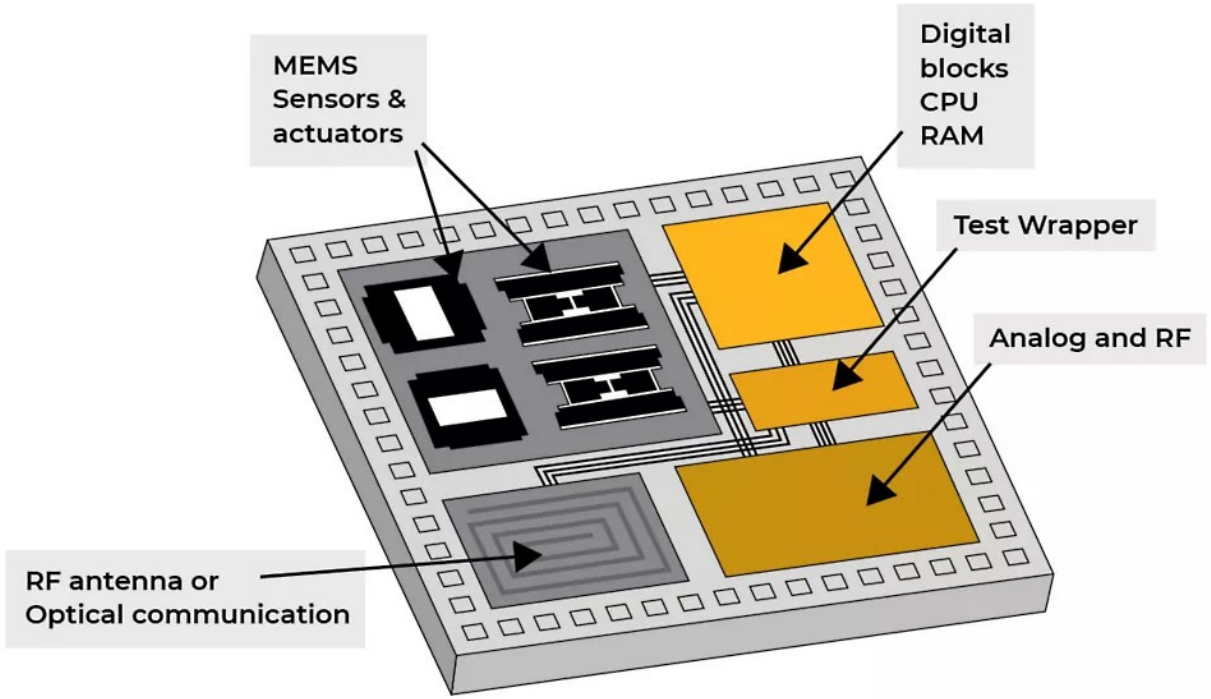
System on Chip – Internal Architecture

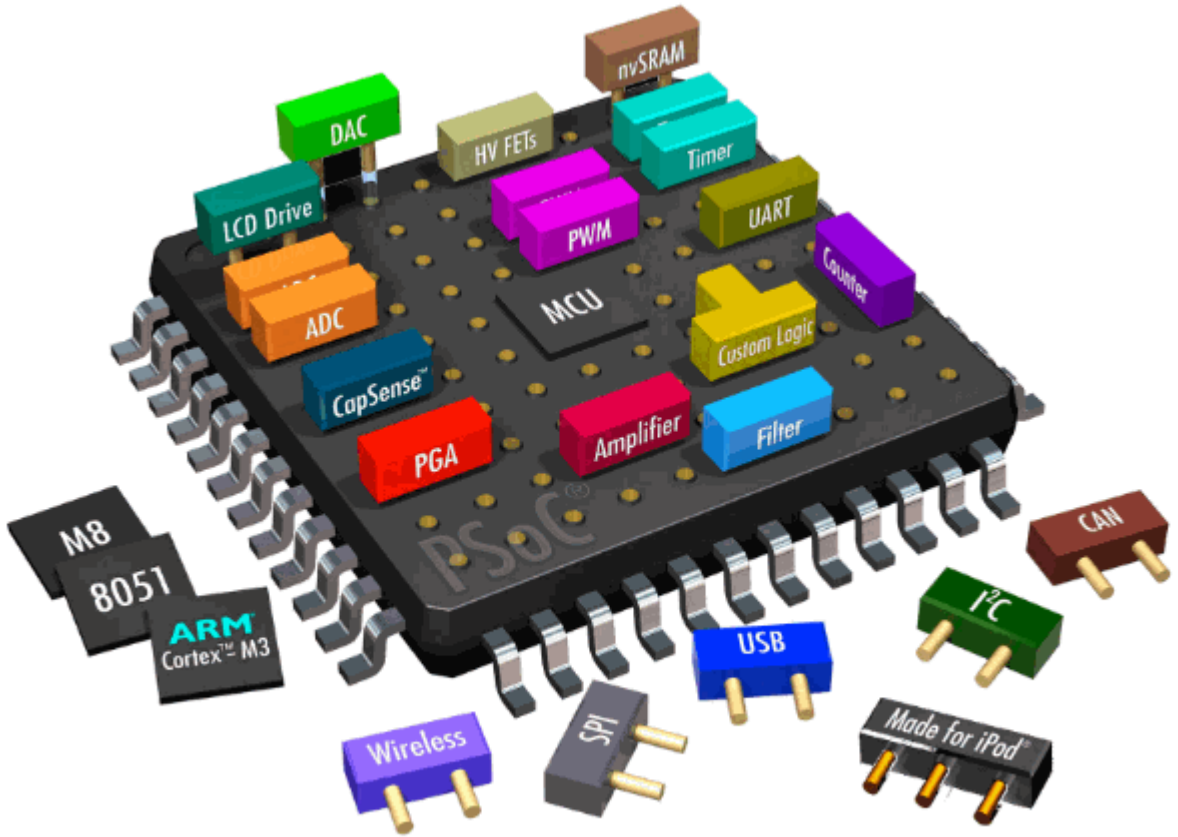


Primary building blocks of SOC are:

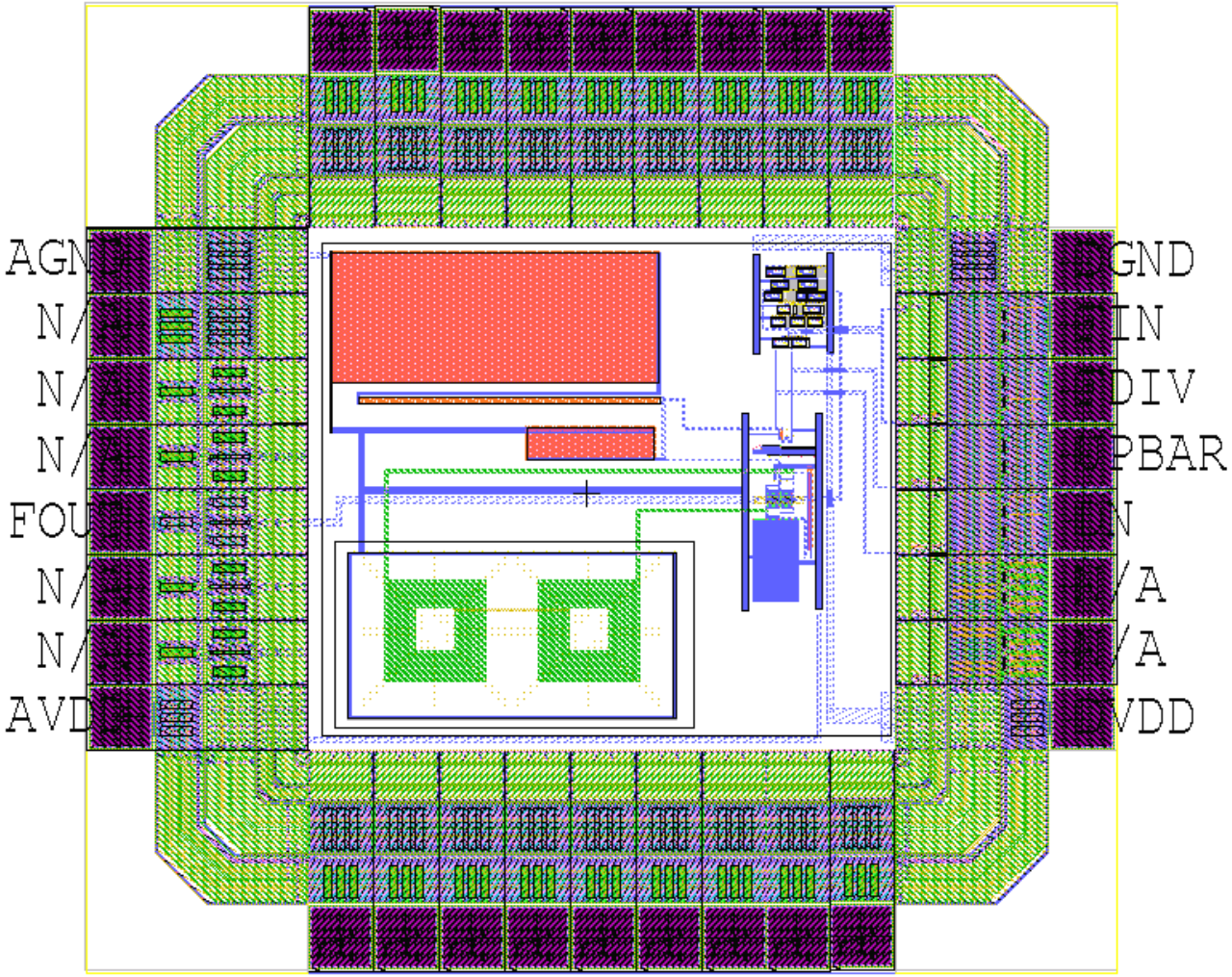
1. Digital
2. Analog
3. Mixed signal

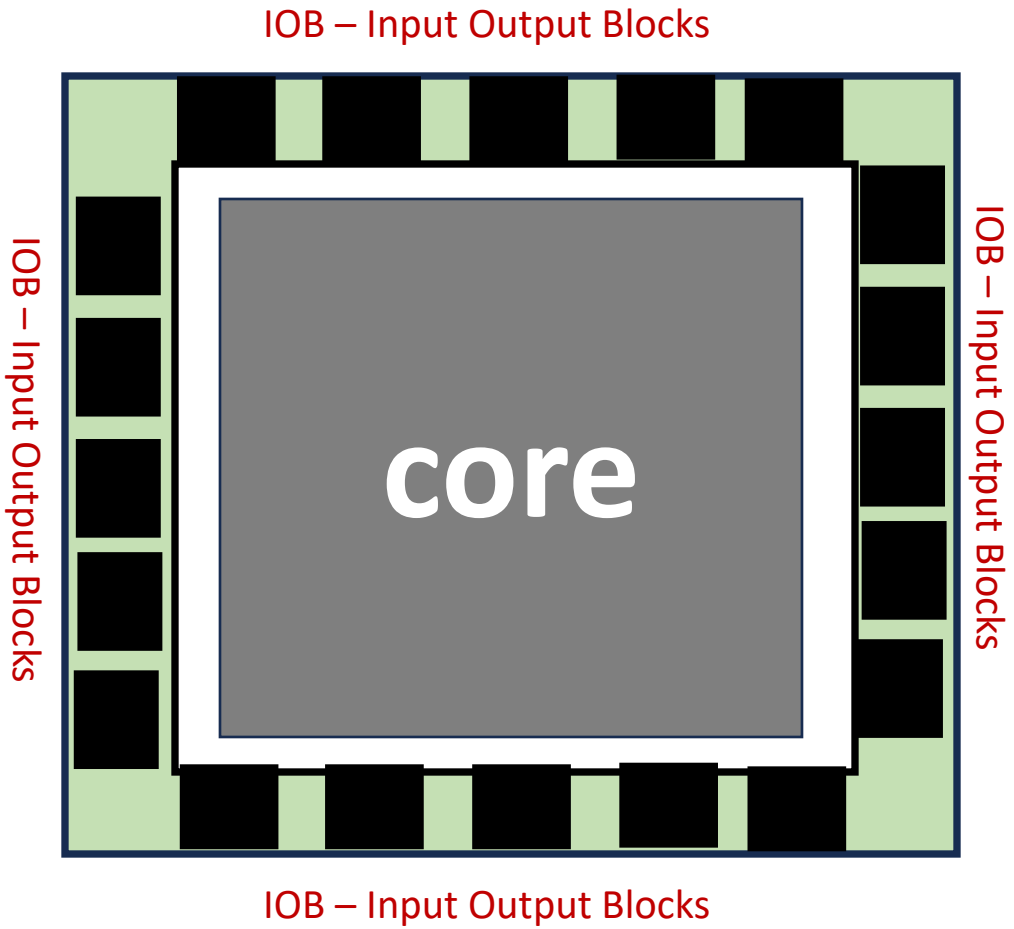
System on Chip





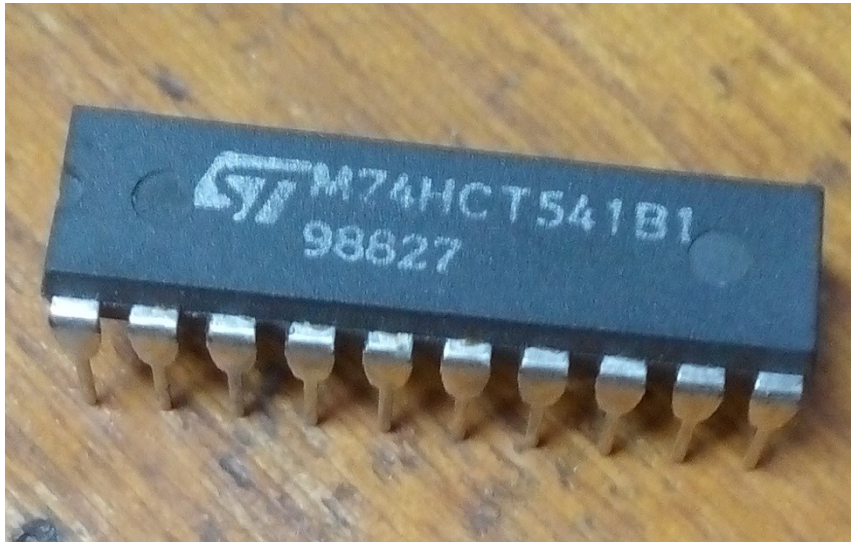
VLSI Full Chip



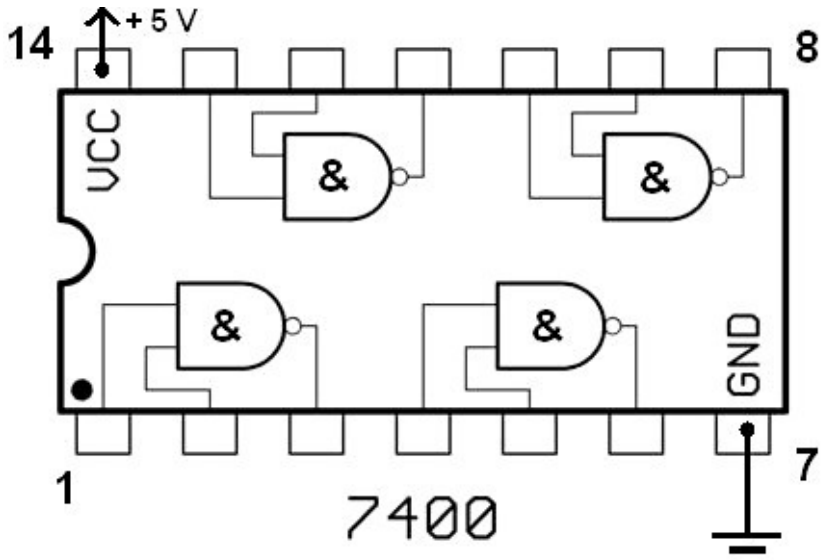


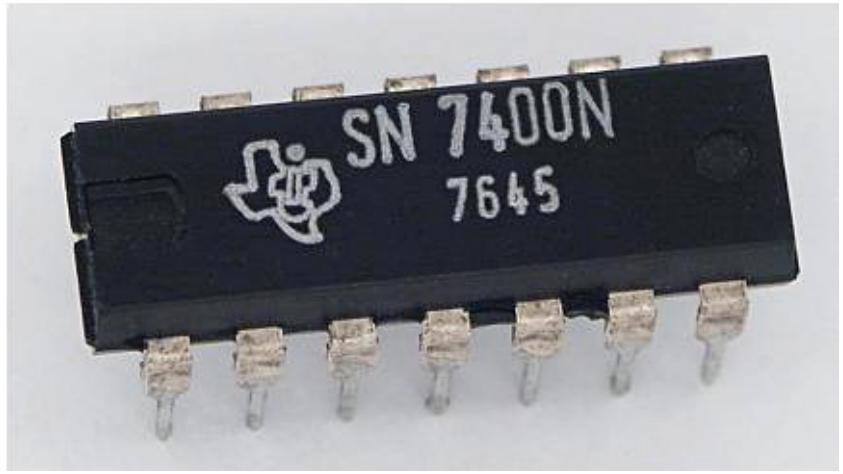
Answer these three questions

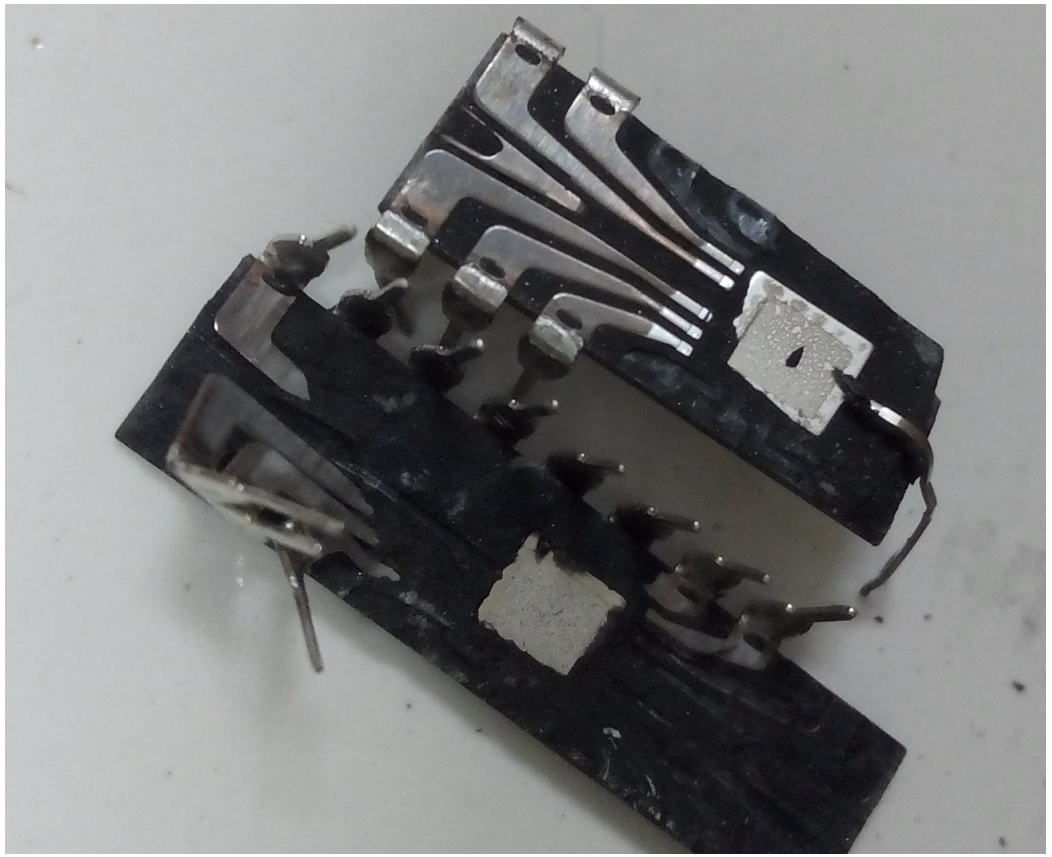
1. Integrated circuit is black (gray) in color ?
   A. Yes or No
2. Integrated circuit has pins on two sides (or four sides) ?
   A. Yes or No
3. The component in the picture is called integrated circuit ?
   A. Yes or No



Integrated circuit

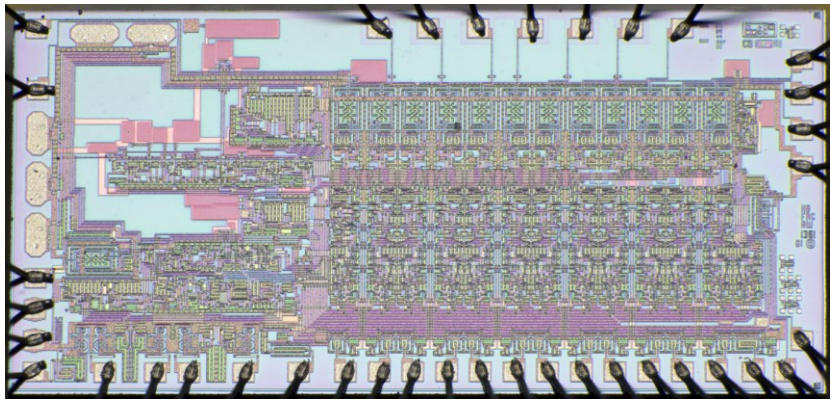


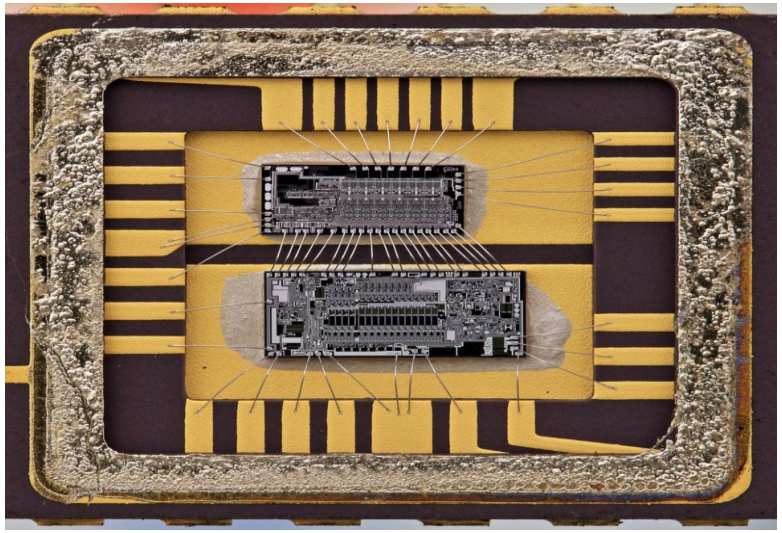


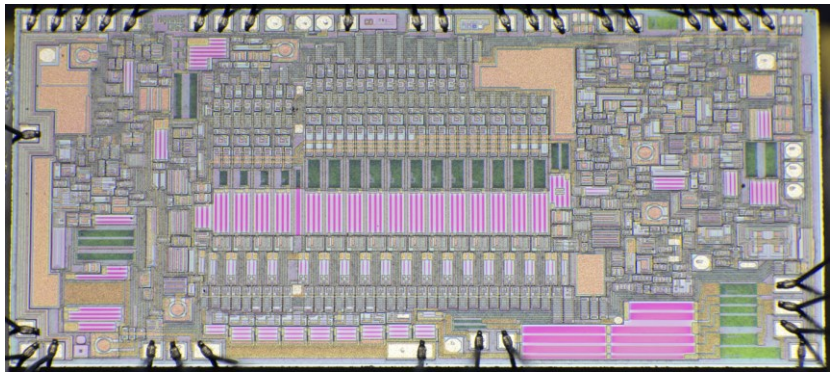


Integrated circuit

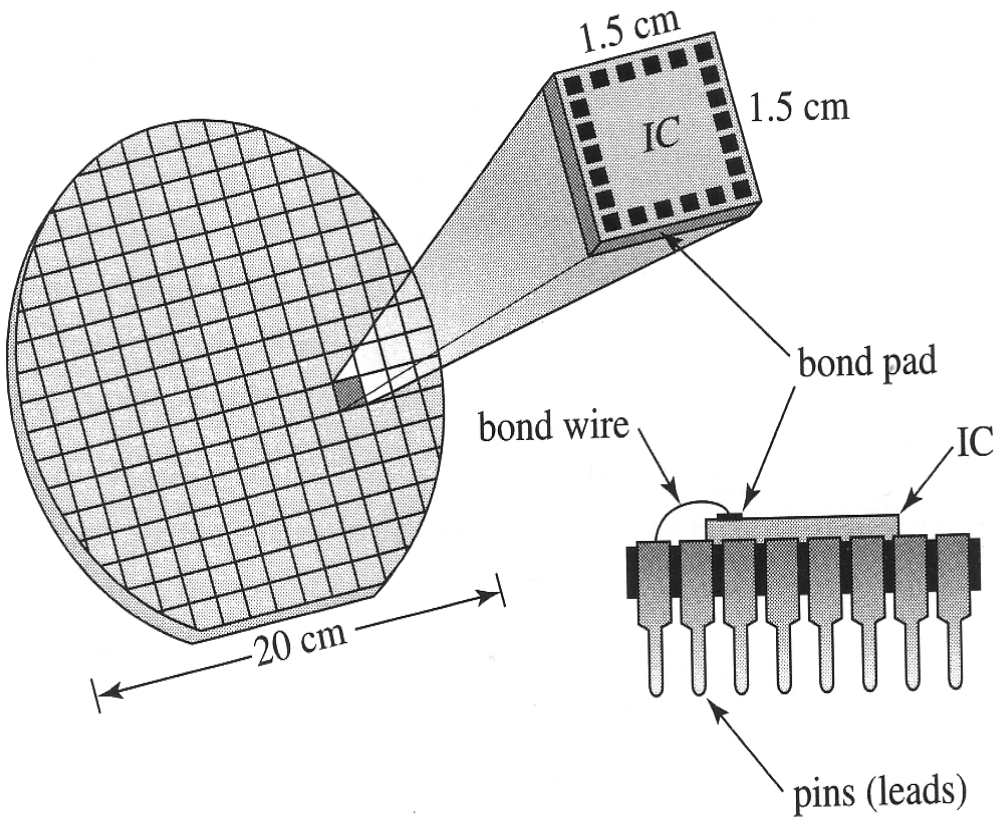


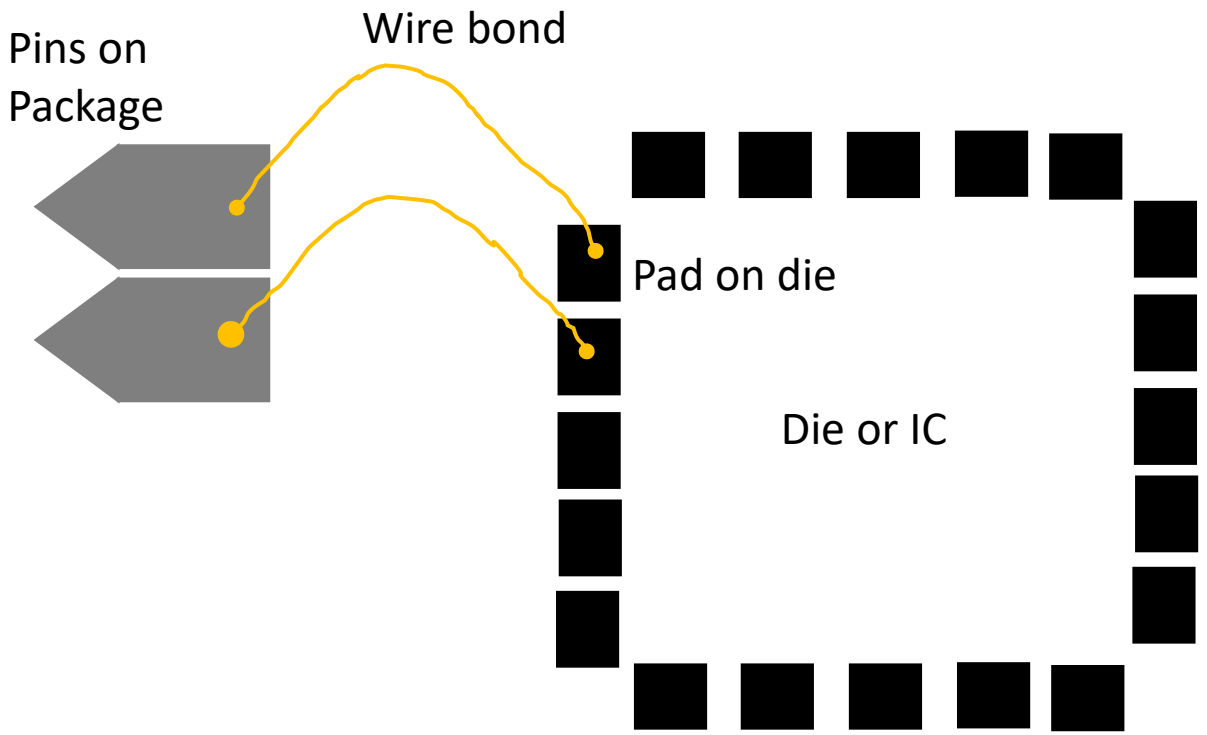


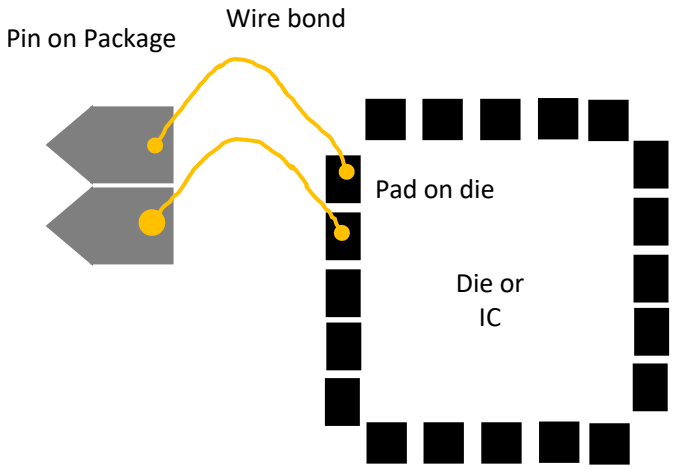


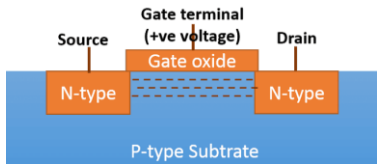


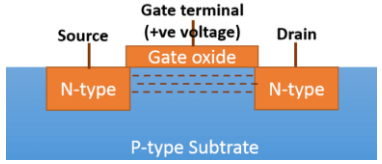
Signal flow in an IC

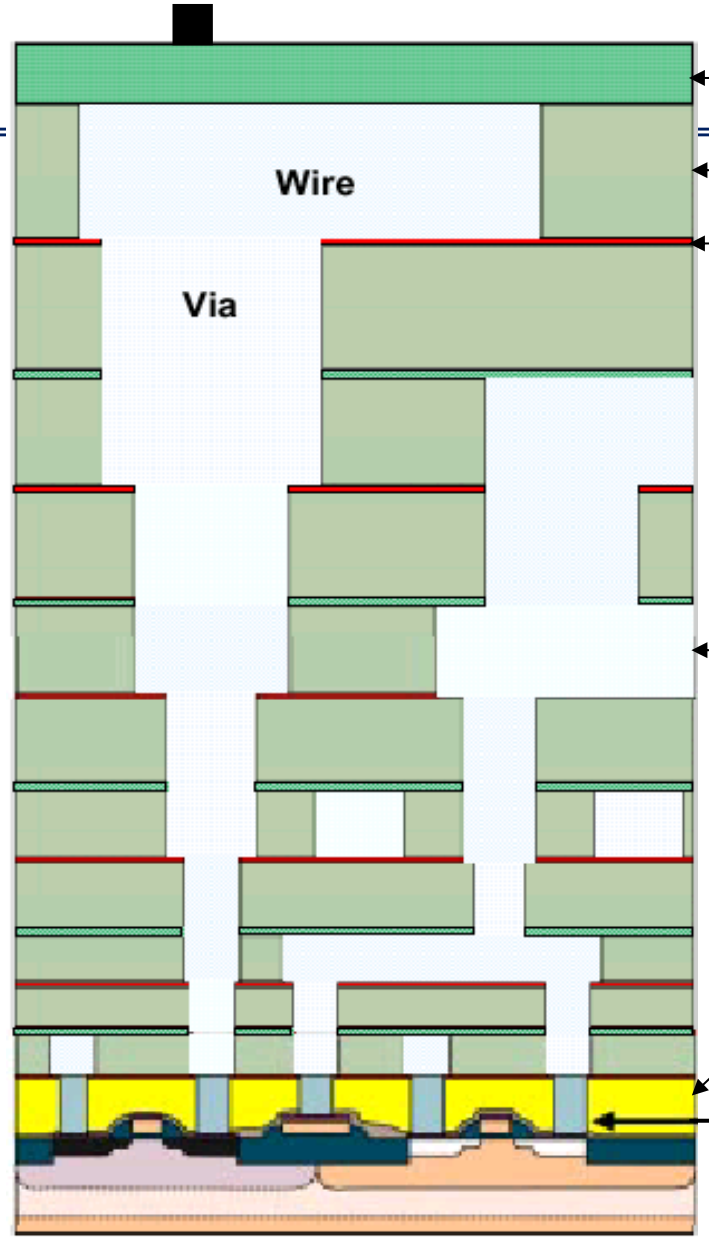


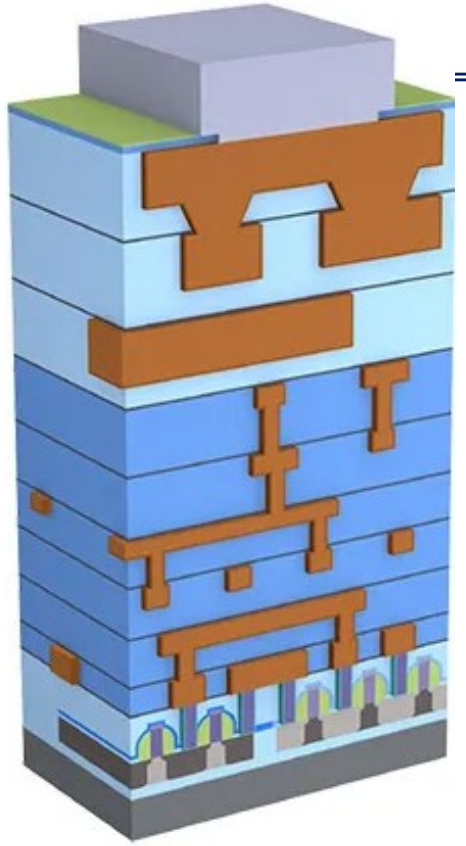


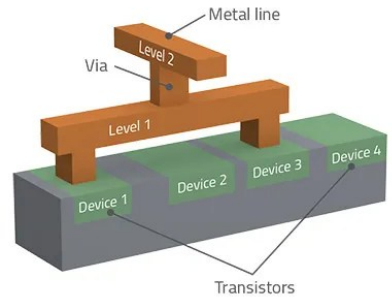






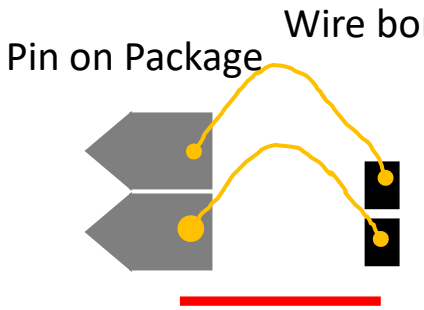




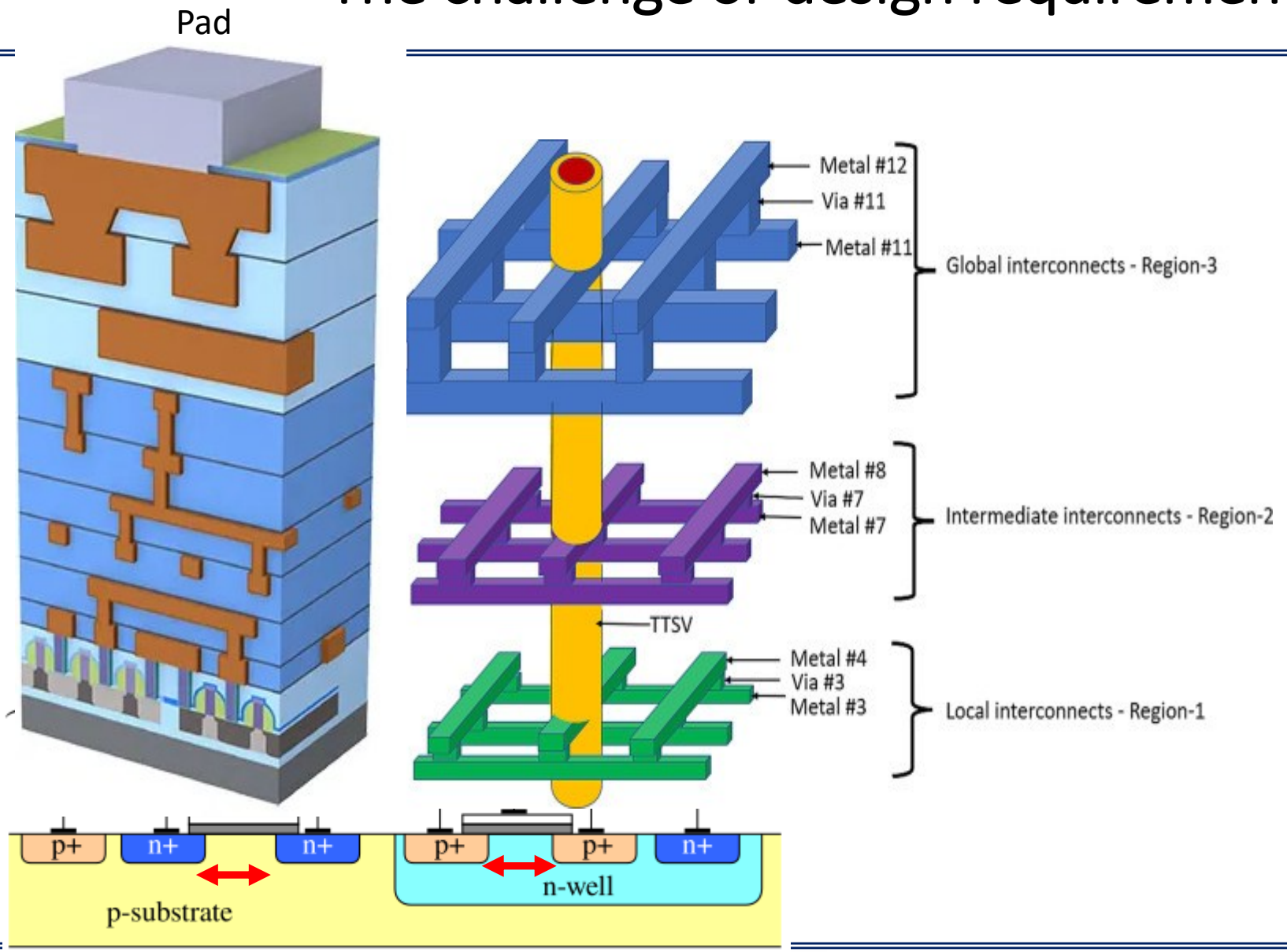


Source: SIA Roadmap 1999

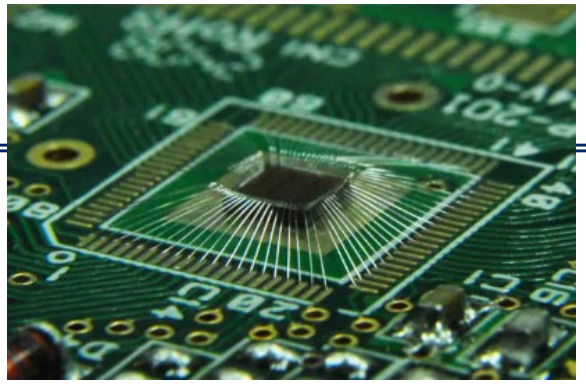
The challenge or design requirement

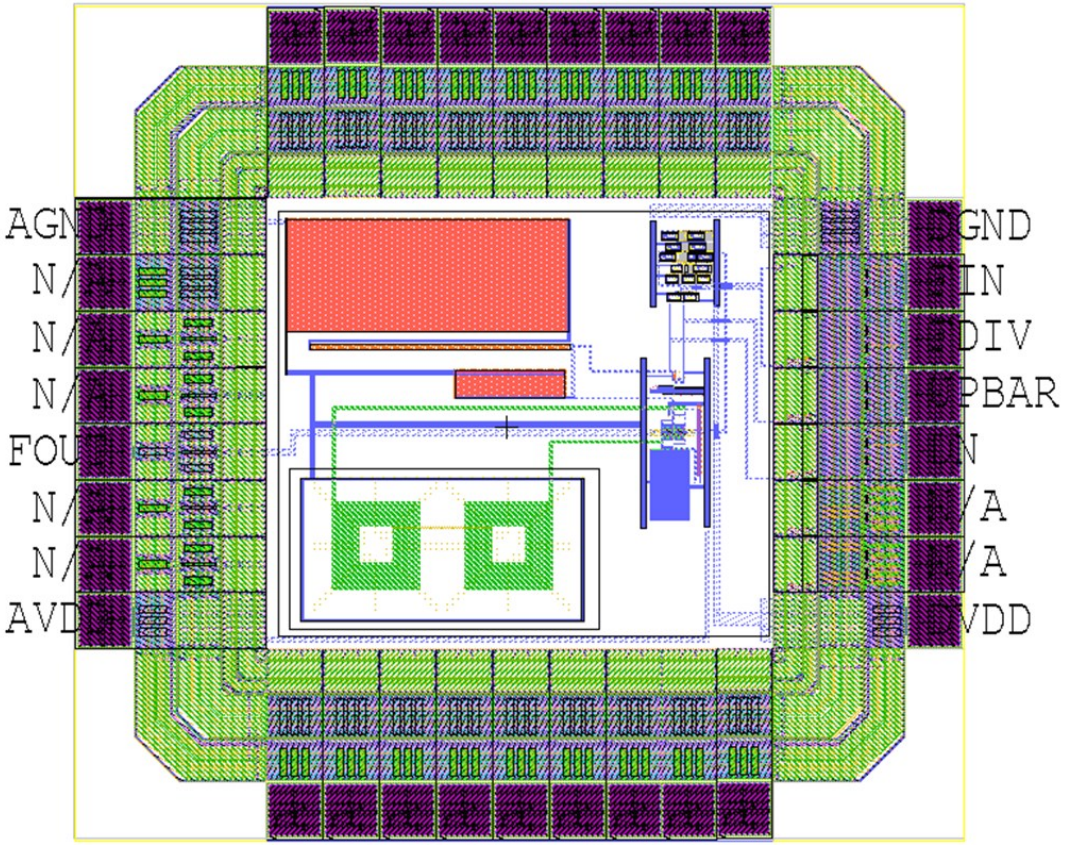


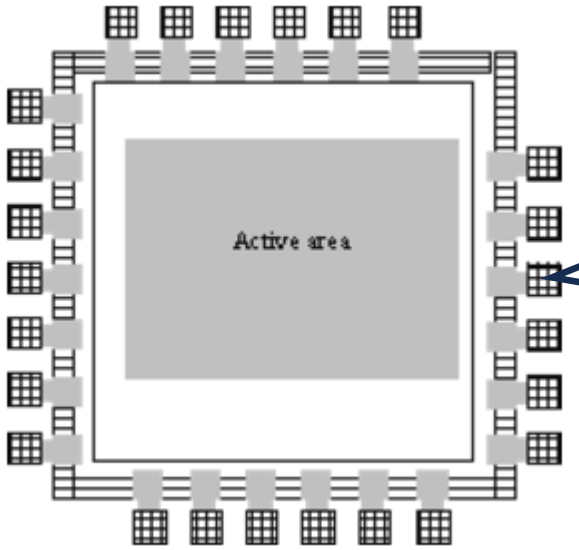
Time taken by signal to reach transistor or switch >> time taken for signal to travel within the channel

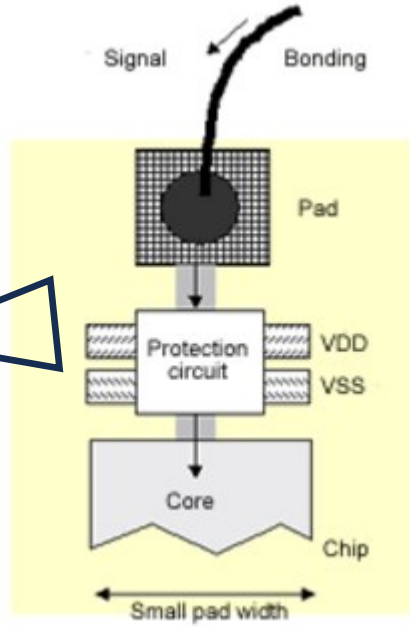


IC and packaging





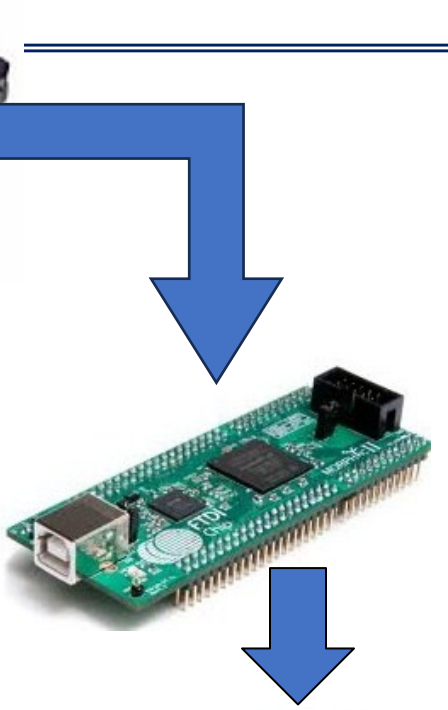


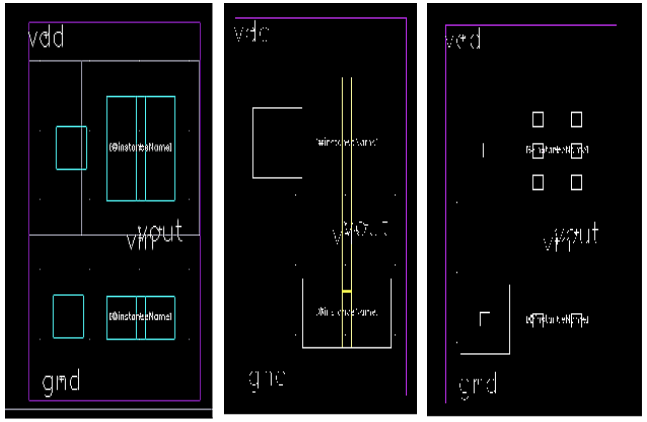


Video Demo of IC Fabrication

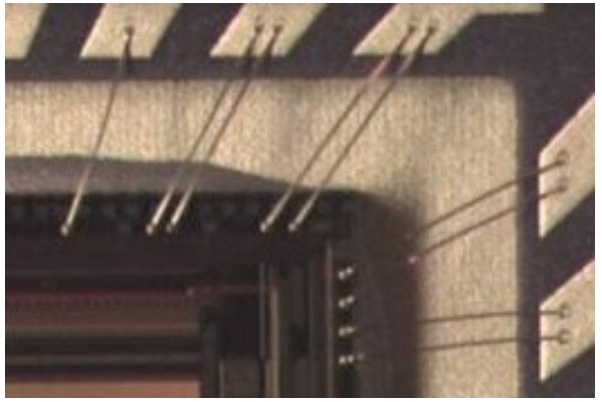
VLSI Design Process

Design Spec. to GDSII → Fabrication → Packaging & Testing

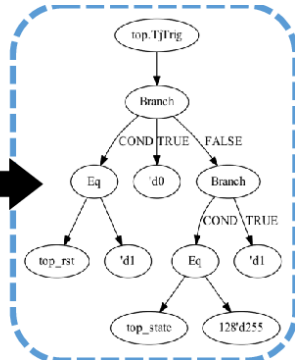


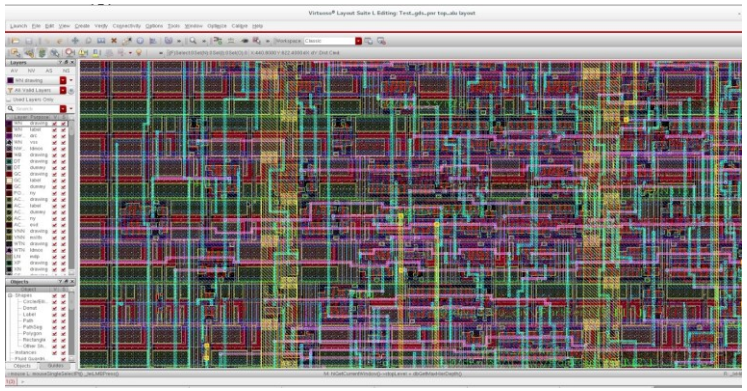






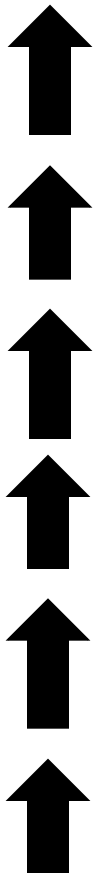
```
module top(
 input rst,
 input [127:0] state,
 output TjTrig
);
 reg TjTrig;
 always @(state) begin
  if (rst == 1)
   TjTrig <= 0;
  else if (state == 128'hFF)
   TjTrig <= 1;
 end
endmodule
```







Full Custom Design



Bottom-Up Approach		
Top Module	Microcontroller	10 MHz, 10 mW, 1 Sq. cm
Components of UC	ALU, MEM, CU	20 MHz
Subsystems of ALU	AU, LU, CU	40 MHz
Subsystems of AU	Adder, Multiplier, Register	80 MHz
Subsystems of Multiplier	Adder, Register, Counter	100 MHz
Subsystems of Adder	Full Adder	120 MHz
Gates in Full Adder	NAND Gate	150 MHz
Transistors of NAND	MOS (PMOS/NMOS)	200 MHz

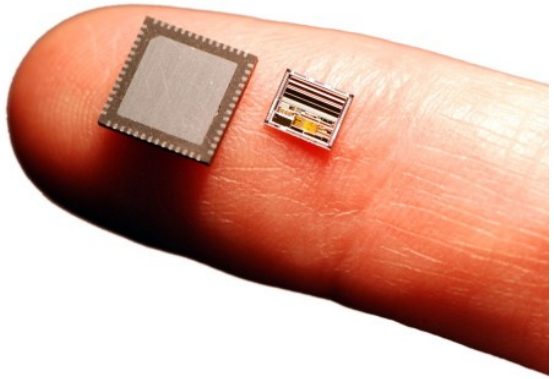
Semicustom Design

- Top Module Microcontroller - 10 MHz, 10 mW, 1 Sq. cm
- Hierarchical Design Approach

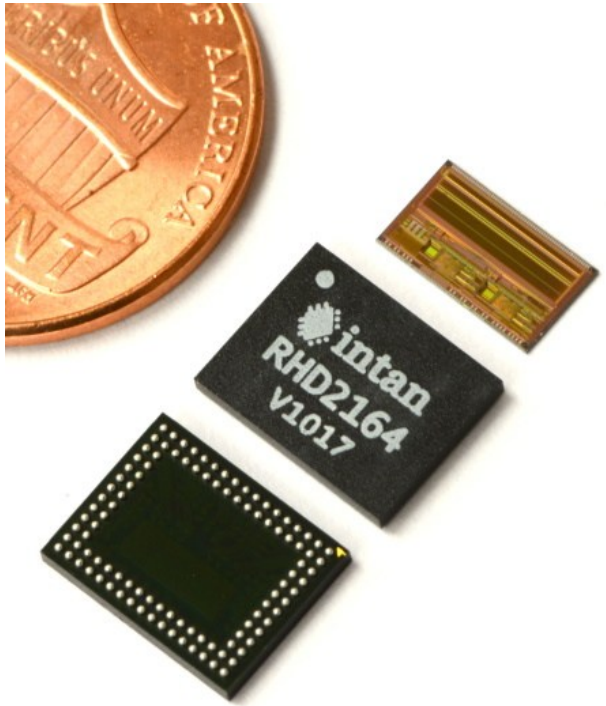
	Team 1	Team 2	Team 3
Sub Modules	ALU	MEM	Control Unit
Functionality	Verilog Code	Verilog Code	Verilog Code
EDA Tool	Verification	Verification	Verification
Set Constraints	Area, Timing, Power	Area, Timing, Power	Area, Timing, Power
EDA Tool	Gate level netlist	Gate level netlist	Gate level netlist
EDA Tool	Verification	Verification	Verification
Integration (EDA tool)	Gate level netlist for Microcontroller		
EDA Tool	VLSI Physical Design		
EDA Tool	Generate GDSII		

Top Down Approach

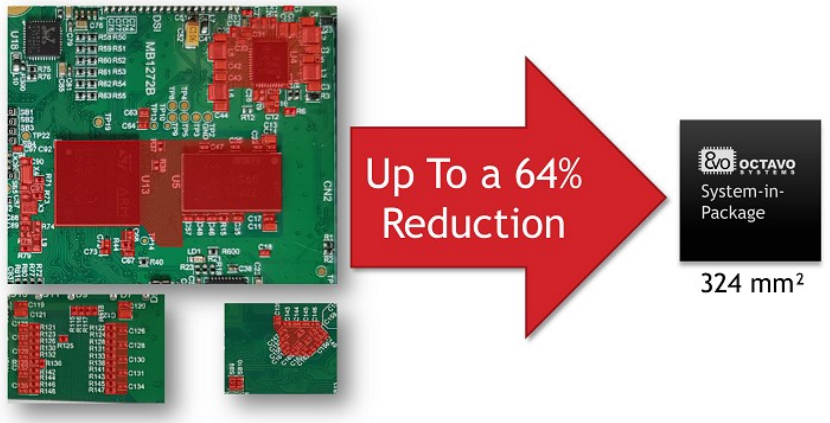
System in Package (SiP)

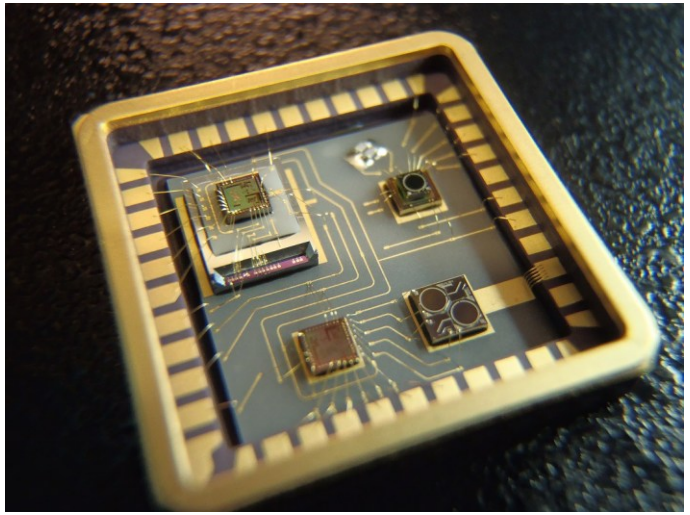


32-channel amplifier chips in QFN package and in bare die form



64-channel amplifier chips in BGA package and in bare die form



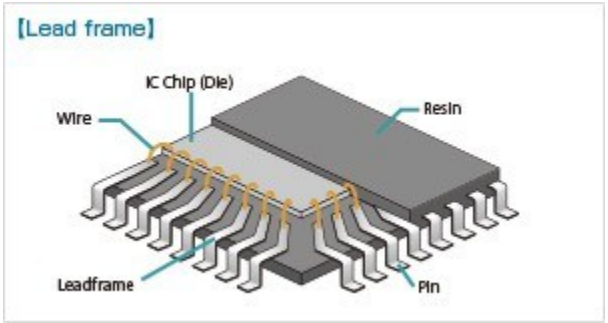


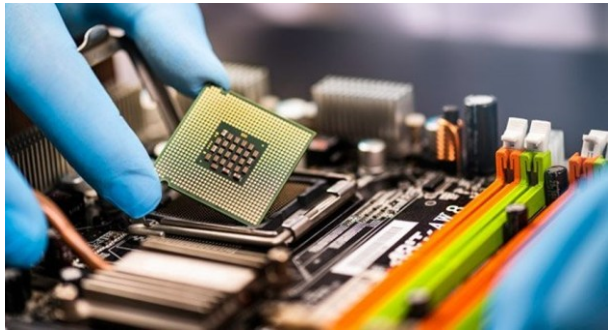
ATMP & OSAT

- Assembly, Testing, Marking, and Packaging (ATMP)

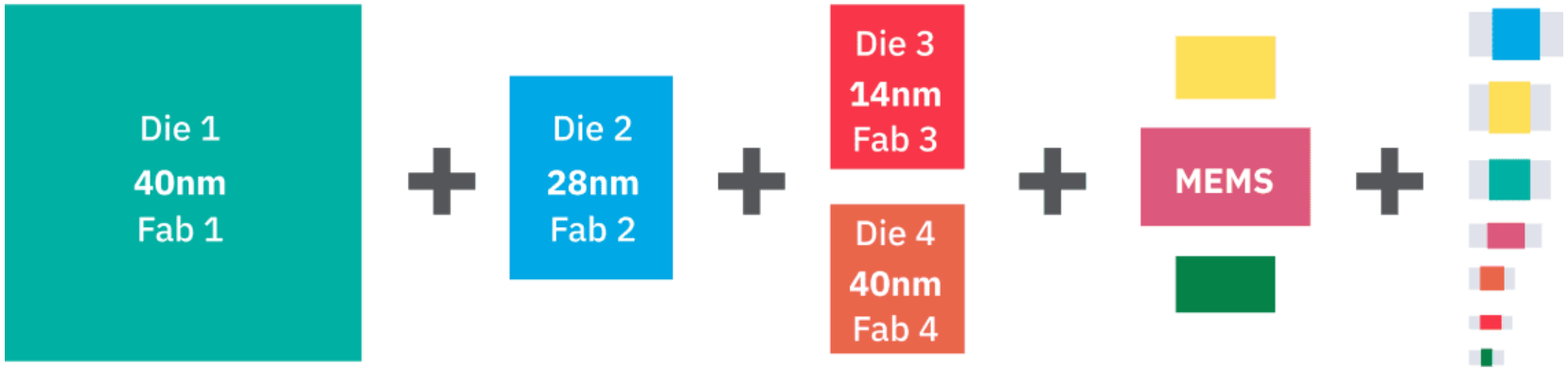


- Outsourced Semiconductor Assembly and Test (OSAT) services



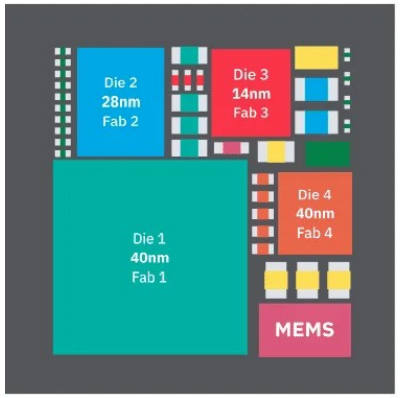


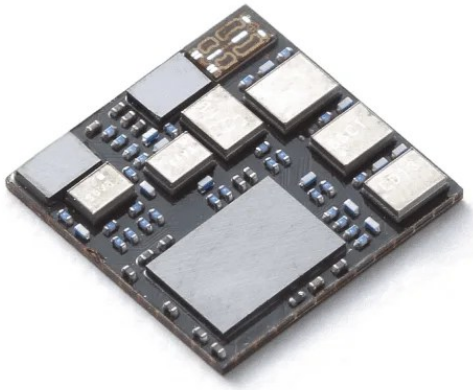
Chiplets: The Next Era of Innovation for the Semiconductor Industry

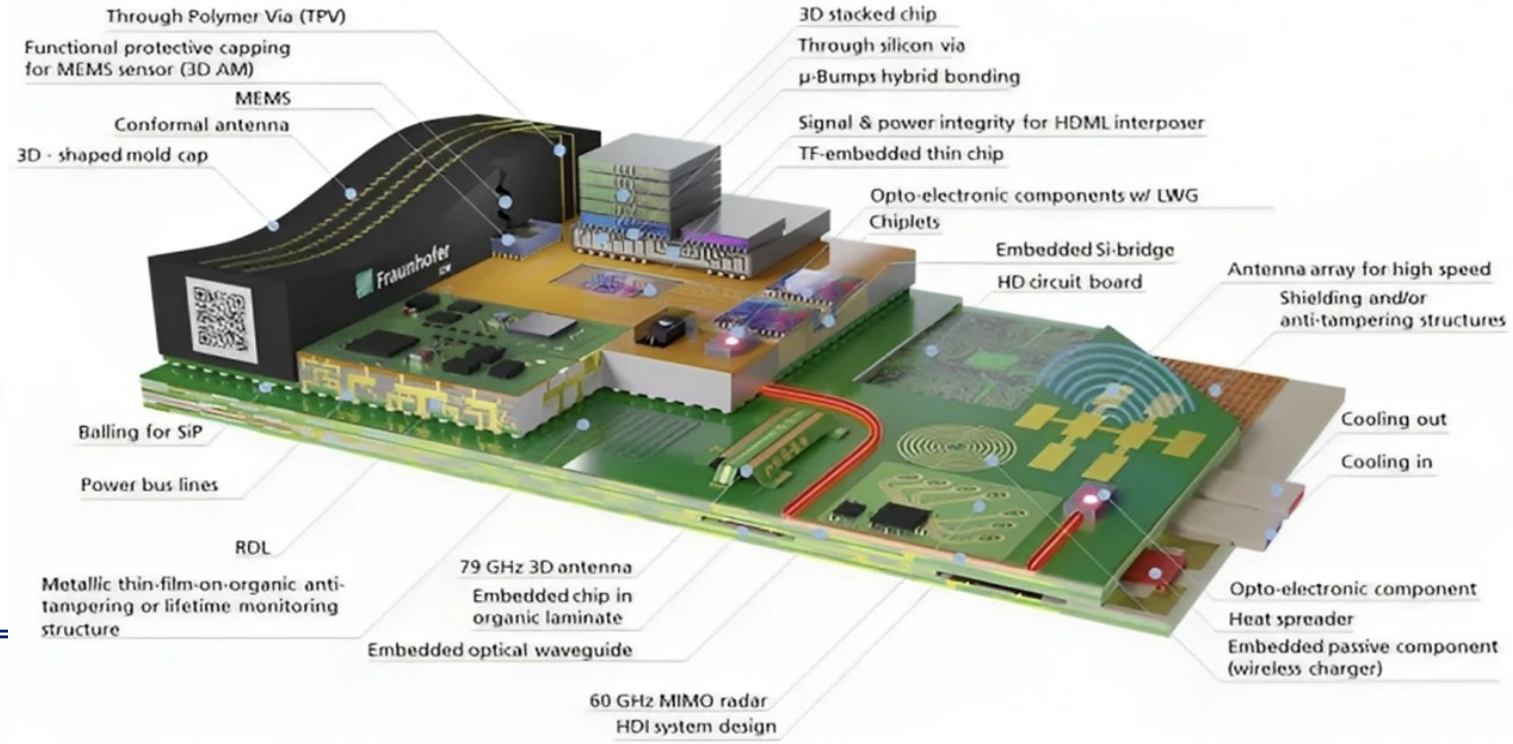


Chiplets represent one of the most promising silicon advancements in decades and offer hope for extending the life of Moore's law

Integration







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CONSULTANCY CENTER AND INDUSTRIAL RESEARCH

VLSI Industry in India



Karnataka Digital Economy Mission

ELECTRONICS SYSTEM DESIGN AND MANUFACTURING

‘Karnataka Will Lead India’s Semiconductor Revolution,’ says State IT/BT Minister Priyank Kharge

Karnataka is positioning itself as a hub for semiconductor innovation and skilling, driving India's transformation from a service-based to a product-driven economy.

The global semiconductor talent shortage

How to solve semiconductor workforce challenges

By 2030, more than one million additional skilled workers will be needed to meet demand in the semiconductor industry. As the competition for talent gets tighter, how can companies address the semiconductor worker shortage and create new possibilities? In this report, learn what’s ahead for the semiconductor supply chain and three actions the industry can take to identify, recruit, and develop the necessary workforce.

Education TIMES

Job Profiles	Skill Set Required	Potential Job Creation by 2026
Chip Semiconductor Fabrication	Fabrication process, equipment maintenance, cleanroom protocols	300,000
Chip Design	Semiconductor design, electronics engineering, CAD tools	200,000
ATMP (Assembly, Testing, Marking, Packaging)	Process engineering, quality assurance, operational knowledge	200,000
Software Development	Embedded systems, firmware development, circuit analysis	120,000
System Circuit	Circuit design, integration, testing, troubleshooting	80,000
Manufacturing Supply Chain Management	Problem solving, data analytics, project management	100,000

TECHNOLOGY

Where India's Semiconductor Mission Stands Today

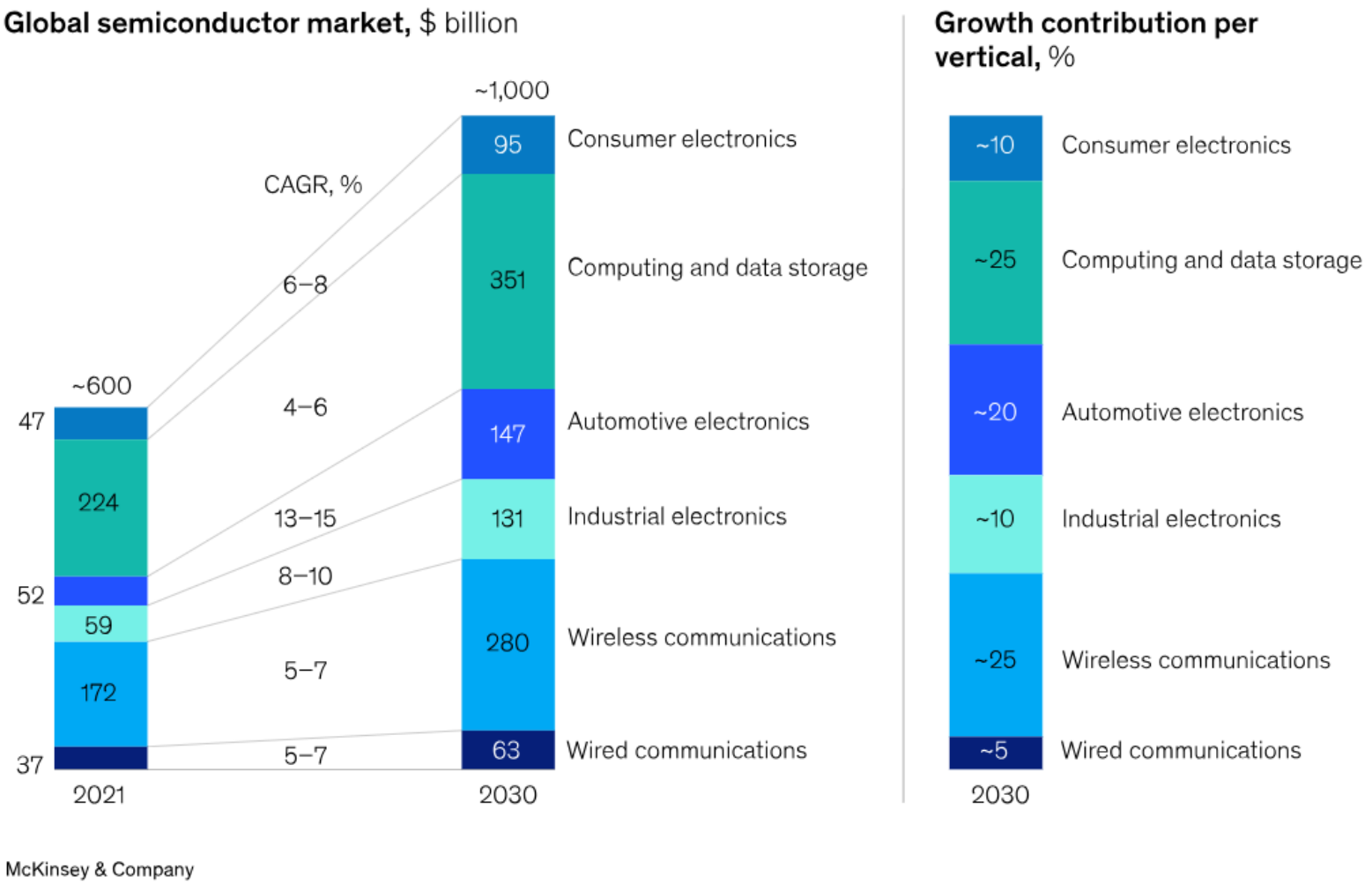
AMIT MISHRA

Sep 23, 2024, 04:00 PM | Updated Sep 27, 2024, 05:10 PM IST

Overview

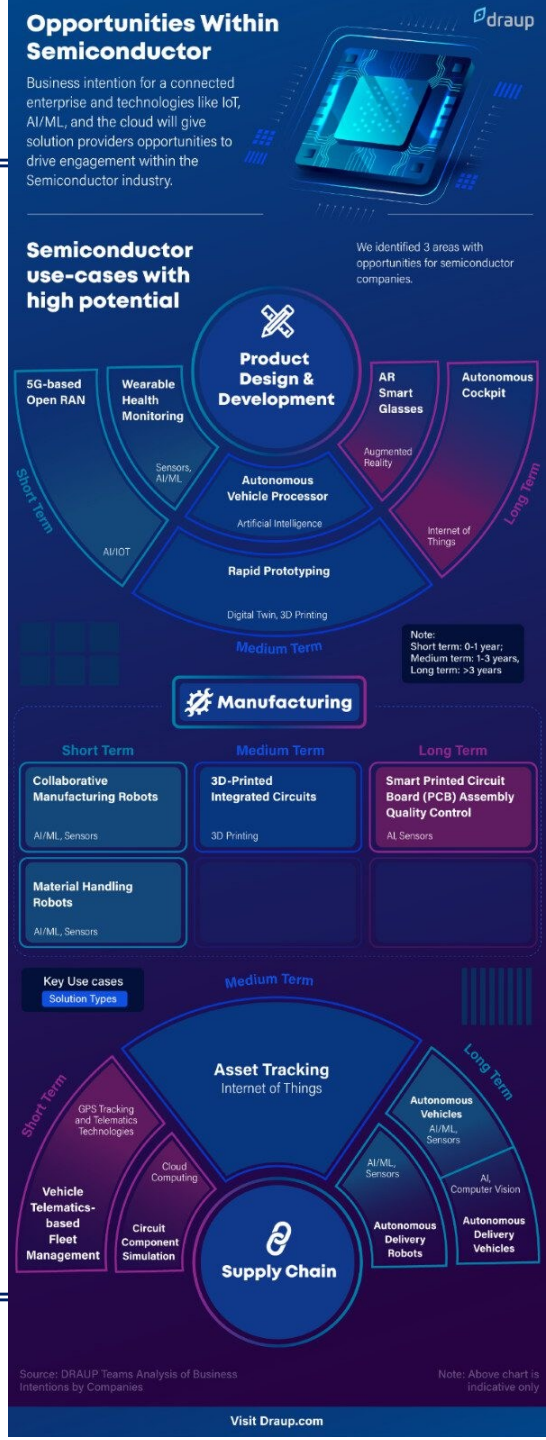
- Total Planned Investment: Approximately Rs 1.50 lakh crore
- Expected Daily Chip Production Capacity: 7 crore chips
- Number of Projects: 7 (including the recently announced India-US joint venture)

Global Semiconductor Market



Trends and Directions for Semiconductor

- **Directions for Semiconductor**
  - **Quantum computing –** qubits revolutionize computation
  - **AI and neuromorphic computing -** mimics the brain to process data
  - **Internet of Things (IoT) and edge computing -** Connected devices ad low power semiconductors
  - **Advanced sensors -** LiDAR, radar, and biosensors
  - **AR and VR applications -** gaming, entertainment, training, and simulation
- **Market Trends and Potential Growth Opportunities**
  - **Electric and autonomous vehicles**
  - **5G communications**
  - **Consumer electronics**
  - **Advanced materials**
  - **Healthcare and medical devices**
  - **Industrial automation**



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CAMBRIDGE INSTITUTE OF TECHNOLOGY An Autonomous Institution

Chips to Startup Programme

Recent Trends – Devices

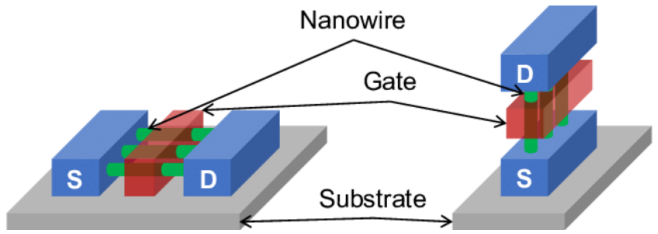
APPLICATIONS DEVELOPMENT

SENSORS INTEGRATION

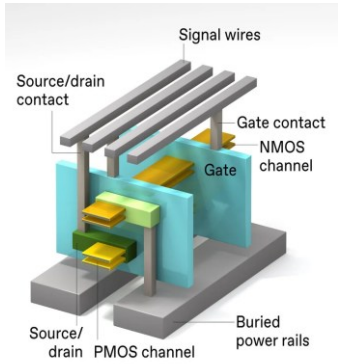
SOC/SIP/HI

CIRCUITS/ SUBSYSTEMS/ SYSTEMS

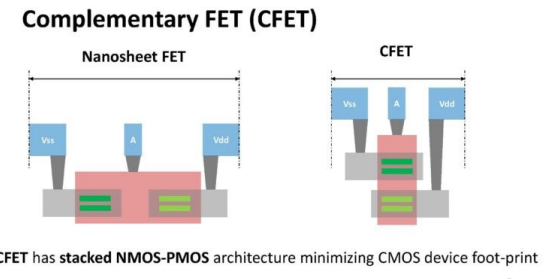
DEVICES



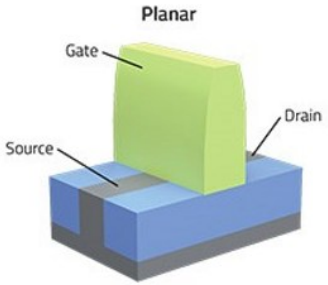
Vertical-channel transistors (VFET)

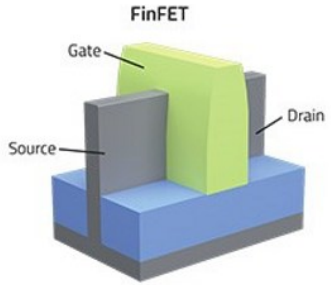


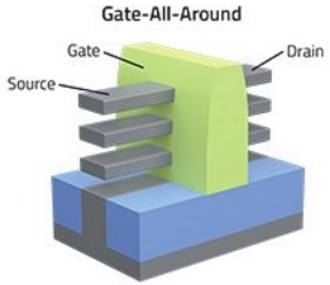
3D stack (3DS) FETs



- fin field-effect transistor (FinFET)
- gate-all-around FET (GAAFET)
- complementary FETs (CFET)
- 3D stack (3DS) FETs
- vertical-channel transistors (VFET)

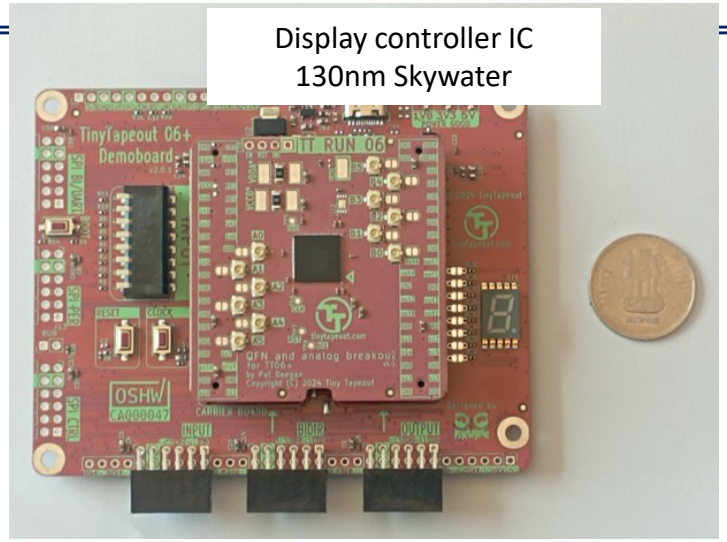


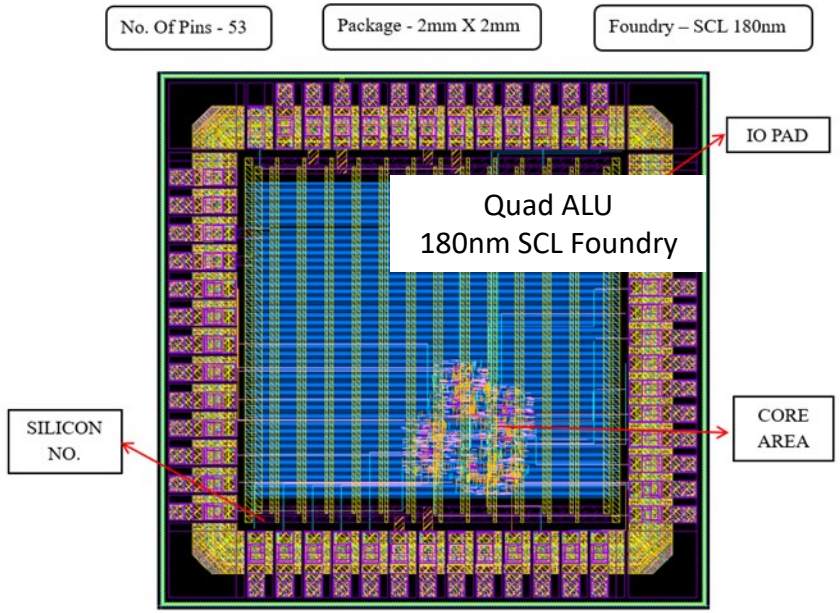


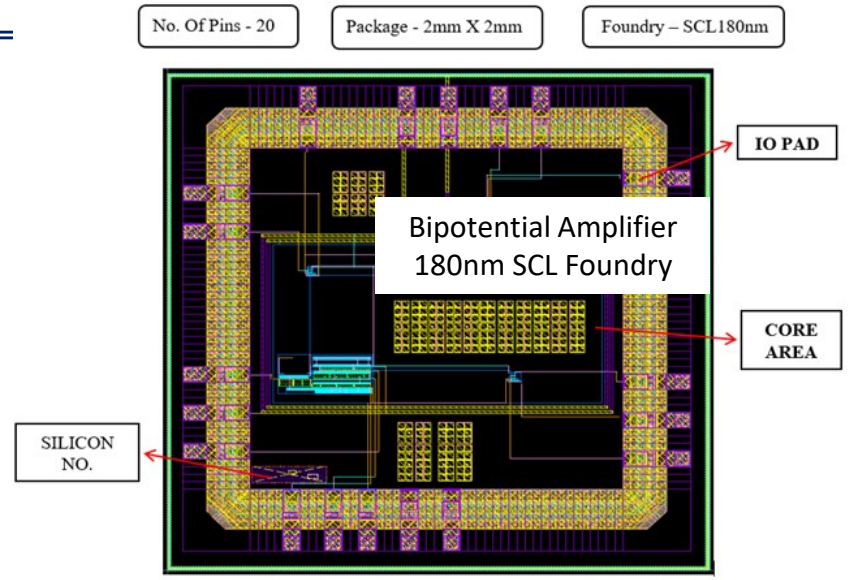


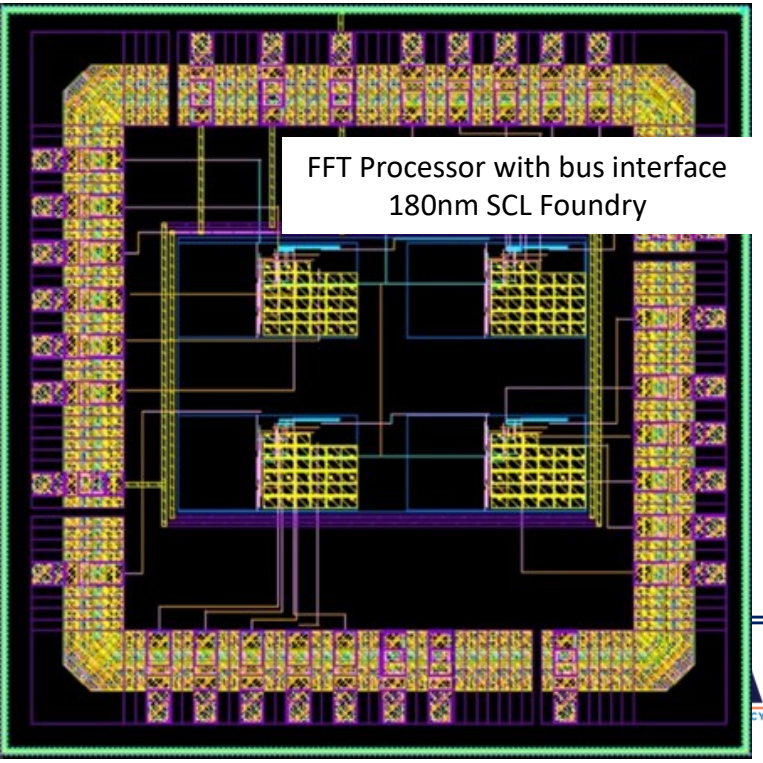


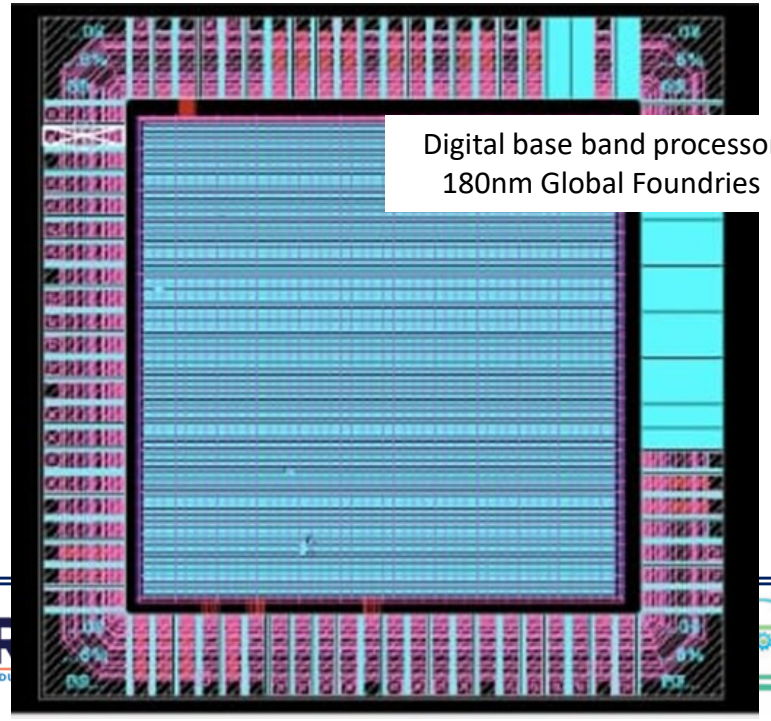
VLSI Chip Tape outs (2025-26)

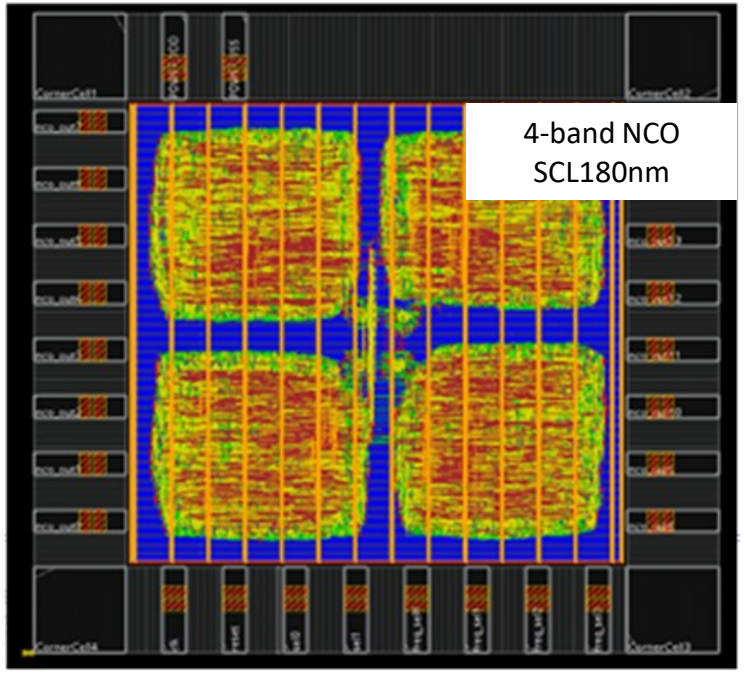




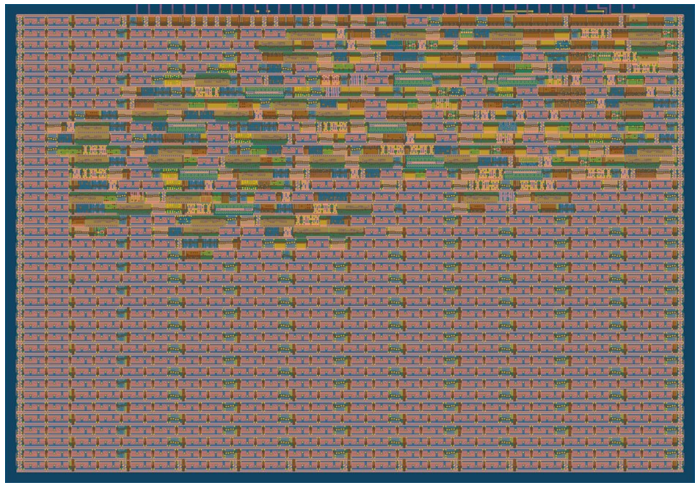




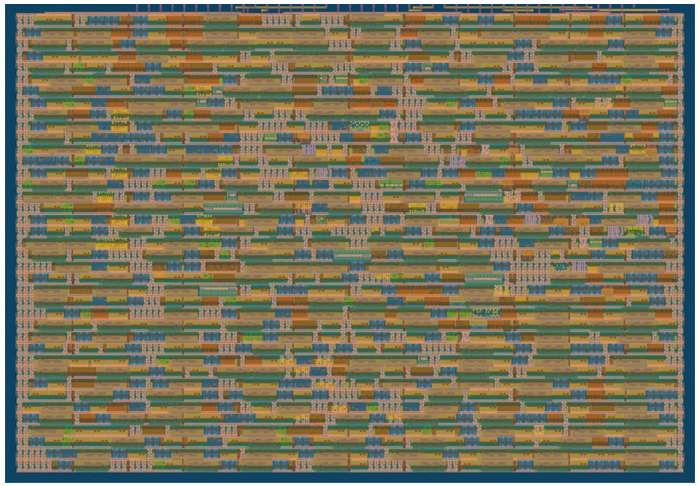




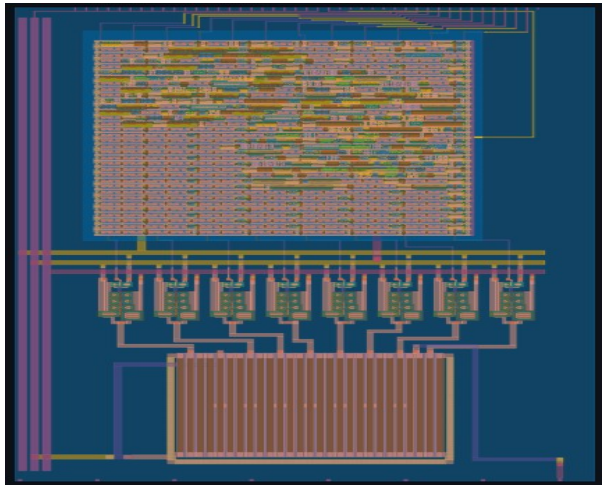
VLSI Chip Tape outs (2026-27)



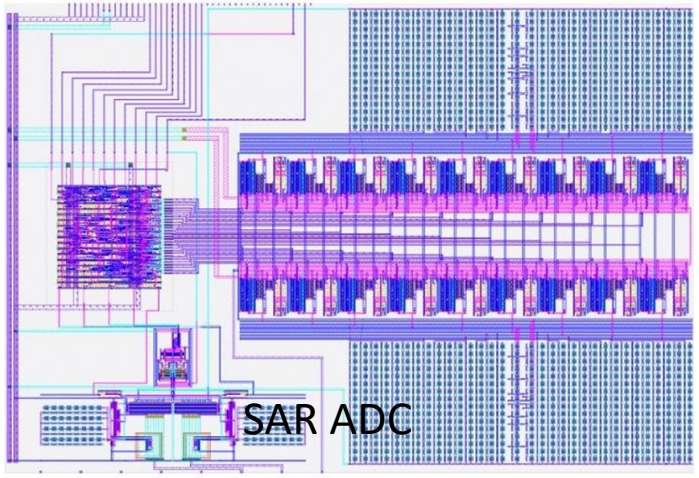
Common False Alarm Rate with 7-segement

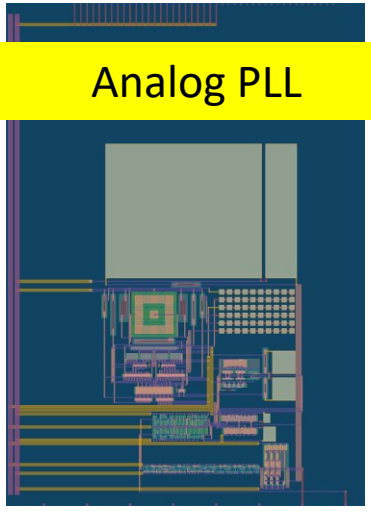


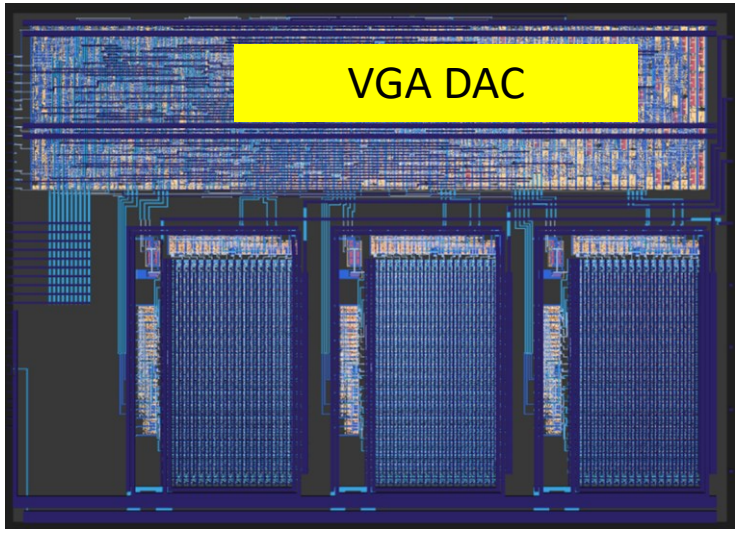
AXI Lite



R2R DAC







25 Tapeouts
25 Silicon IPs
25 Patents/ IPRs
5 Startups
80 student employed

What Are the Future Trends in Analog and Mixed-Signal (AMS) Design?

Importance of Analog Design

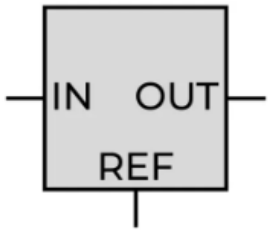
- Typical objectives of analog design include:
  - Signal fidelity
  - Amplification
  - Filtering
- Analog design must compensate for all of these effects to ensure three basic qualities: fidelity/precision, consistency, and performance.

Analog Design vs. Digital Design

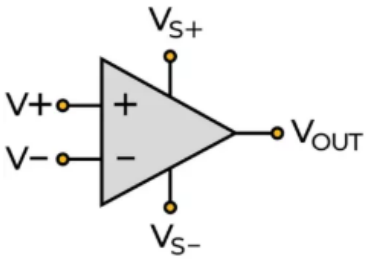
Attribute	Analog Design	Digital Design
Signal Type	Continuous signals	Discrete signals (logic "1" and "0")
Primary Focus	Fidelity, precision, consistency, and performance of waveforms	Logical correctness and timing of binary states
Analysis Domain	Time and frequency domain	Combinatorial and sequential logic models
Variability Handling	Compensates for manufacturing and operational variability (voltage, temperature, interaction effects)	Relies on consistent circuit behavior at defined voltage levels
Design Goal	Accurate sensing, amplification, filtering, and signal integrity	Reliable processing of logical operations
Simulation Method	Circuit simulation with linear and nonlinear models, parasitic-aware	Logic simulation using abstracted binary models
Layout Considerations	Requires careful layout for parasitic effects, density, and manufacturability	Layout focused on timing closure, area efficiency, and functional correctness
Common Applications	IoT sensors, RF circuits, analog front ends, precision amplifiers	Microprocessors, digital controllers, memory, logic blocks
Verification	Parasitic extraction, signal integrity, and LVS (layout versus schematic) checking	Functional verification, formal equivalence checking, static timing analysis

Examples and Applications of Analog ICs

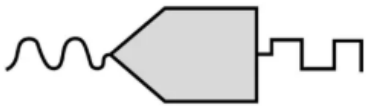
Voltage and current regulators: It is commonly used in power circuits where the output should remain constant irrespective of varying input voltage. The regulator IC ensures constant output. Voltage regulators are commonly used in all power supply systems.



Operational Amplifiers (op-amp): Op-amps are used to amplify the input signal to a higher level as required by the load. Op-amps can also filter unwanted signals. The most common application of op-amp ICs is audio amplifiers.

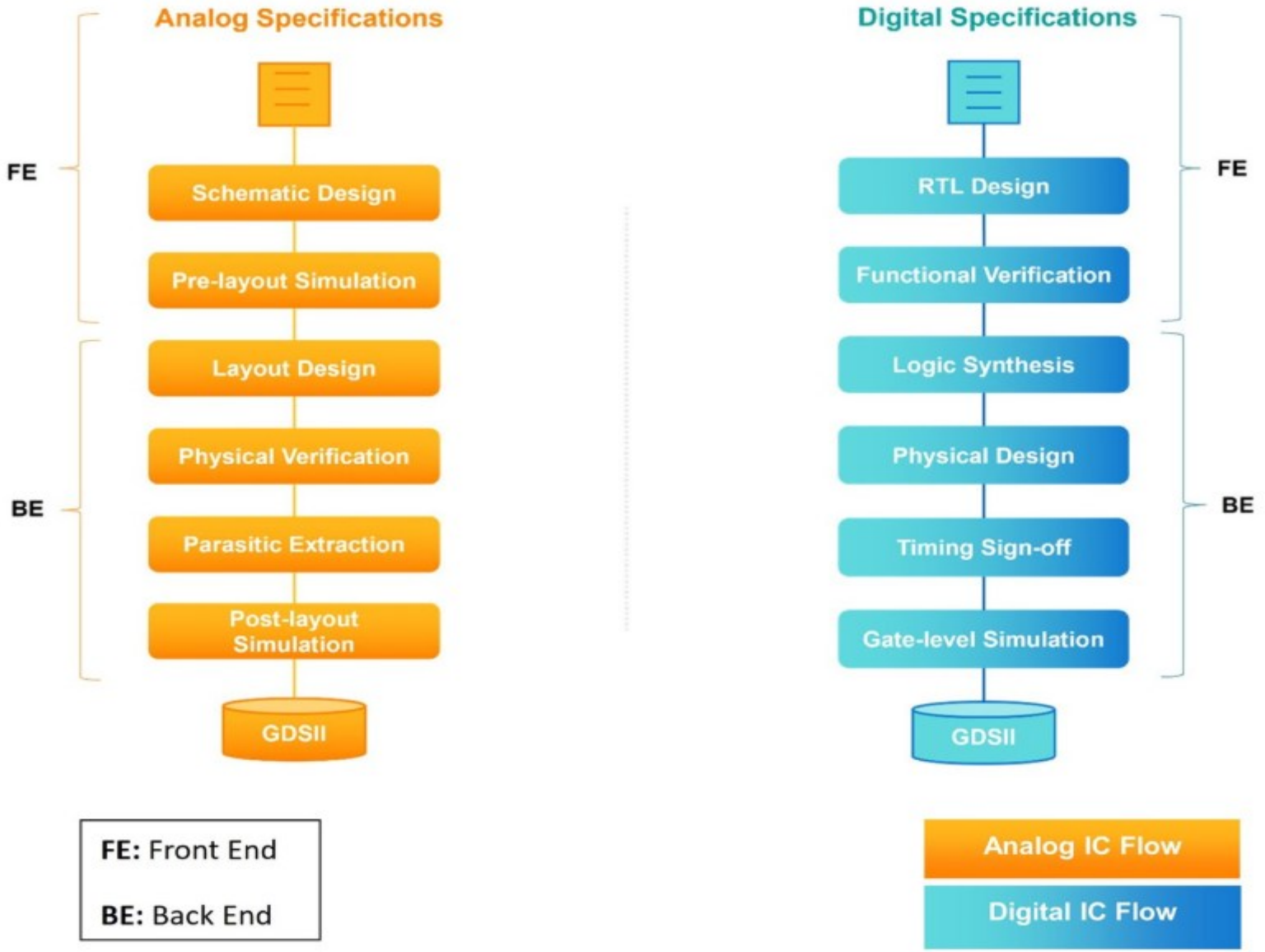


Data converter ICs: These ICs are used to convert analog signals into digital signals and are known as Analog-digital-converters (ADCs). These ICs are used where the input signal is continuous (sound, heat, etc.) and has to be converted into a digital stream for processing. The common application is radio.

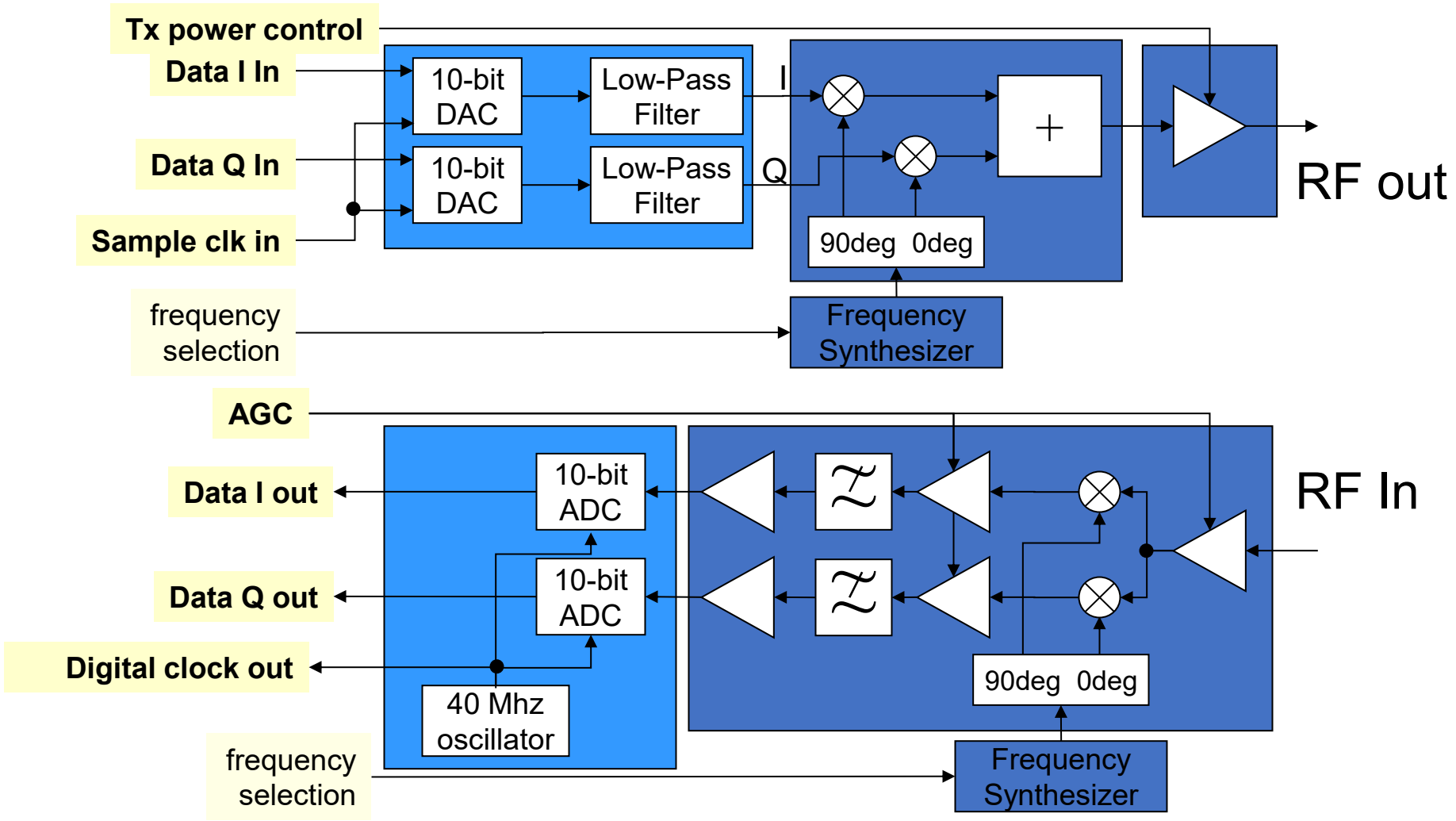


Audio amplifiers: Audio amplifier ICs are used in radiofrequency applications where the received analog signal is very weak. They accept the low input frequency and raise it to a higher level that is required by the circuit. These chips are commonly used in communication systems.

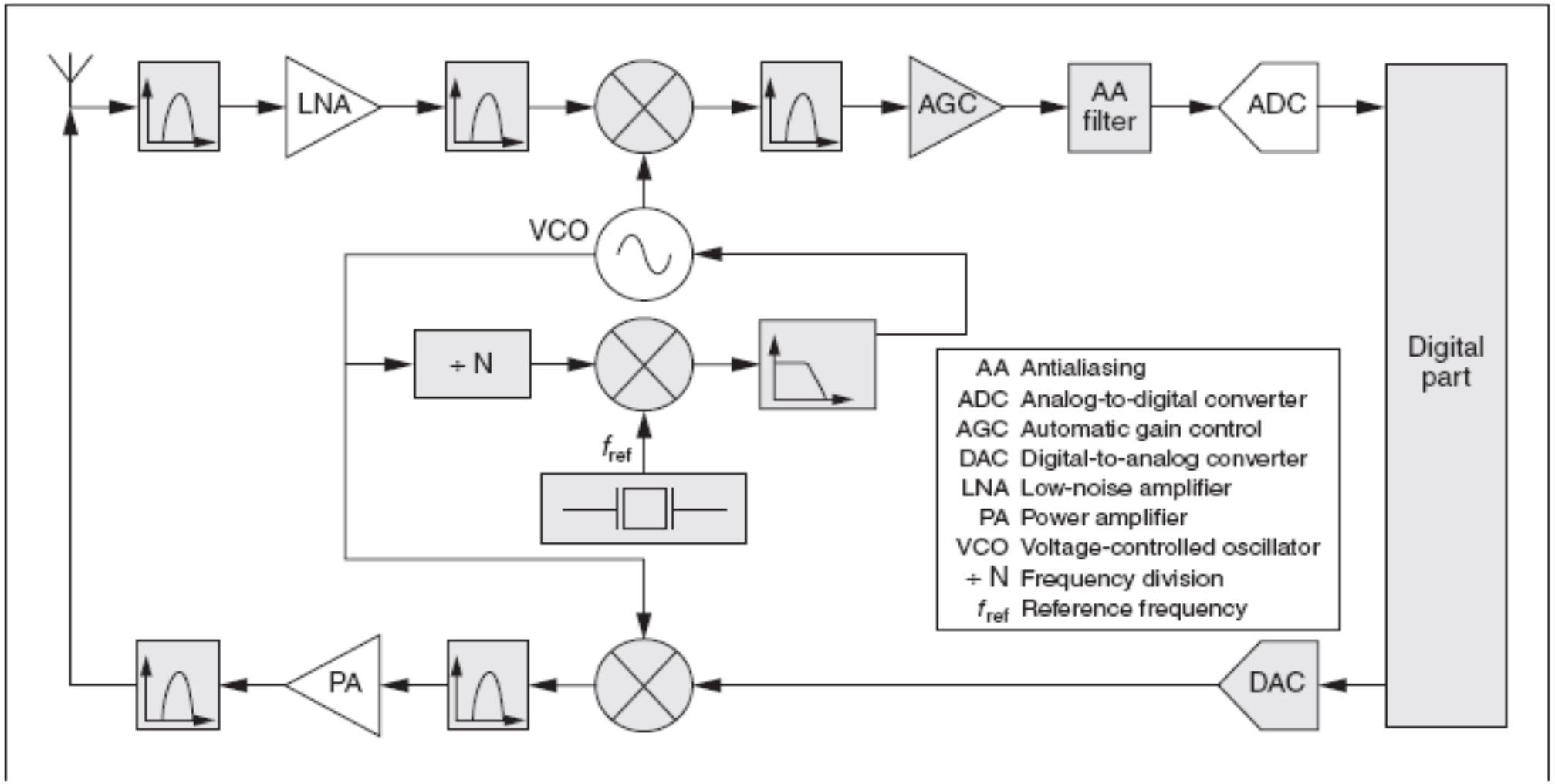



Analog Specifications
FE
Schematic Design
Pre-layout Simulation
BE
Layout Design
Physical Verification
Parasitic Extraction
Post-layout Simulation
GDSII
Digital Specifications
FE
RTL Design
Functional Verification
BE
Logic Synthesis
Physical Design
Timing Sign-off
Gate-level Simulation
GDSII
FE: Front End
BE: Back End
Analog IC Flow
Digital IC Flow

RF Software Radio Hardware



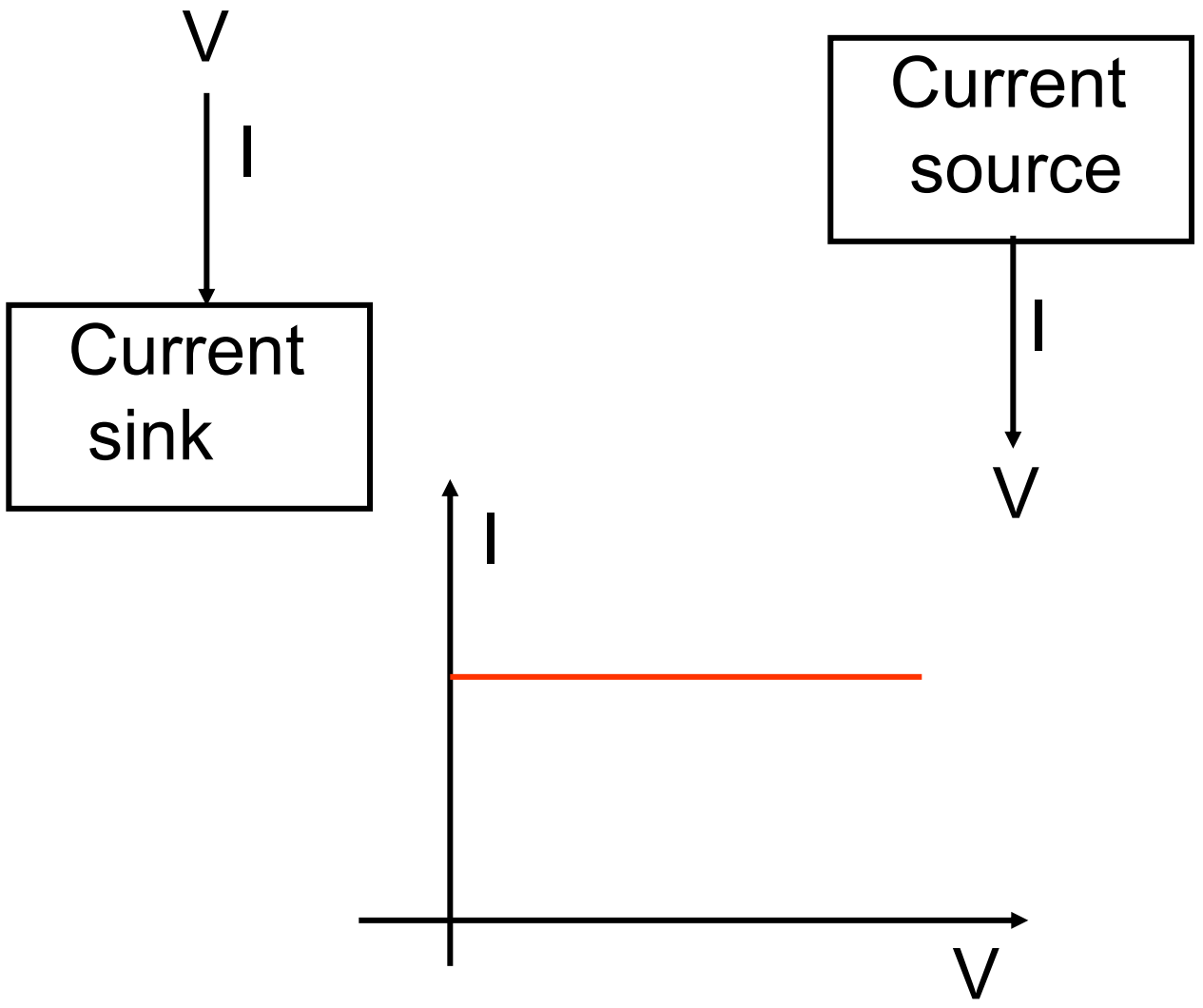
Mixed signal IC



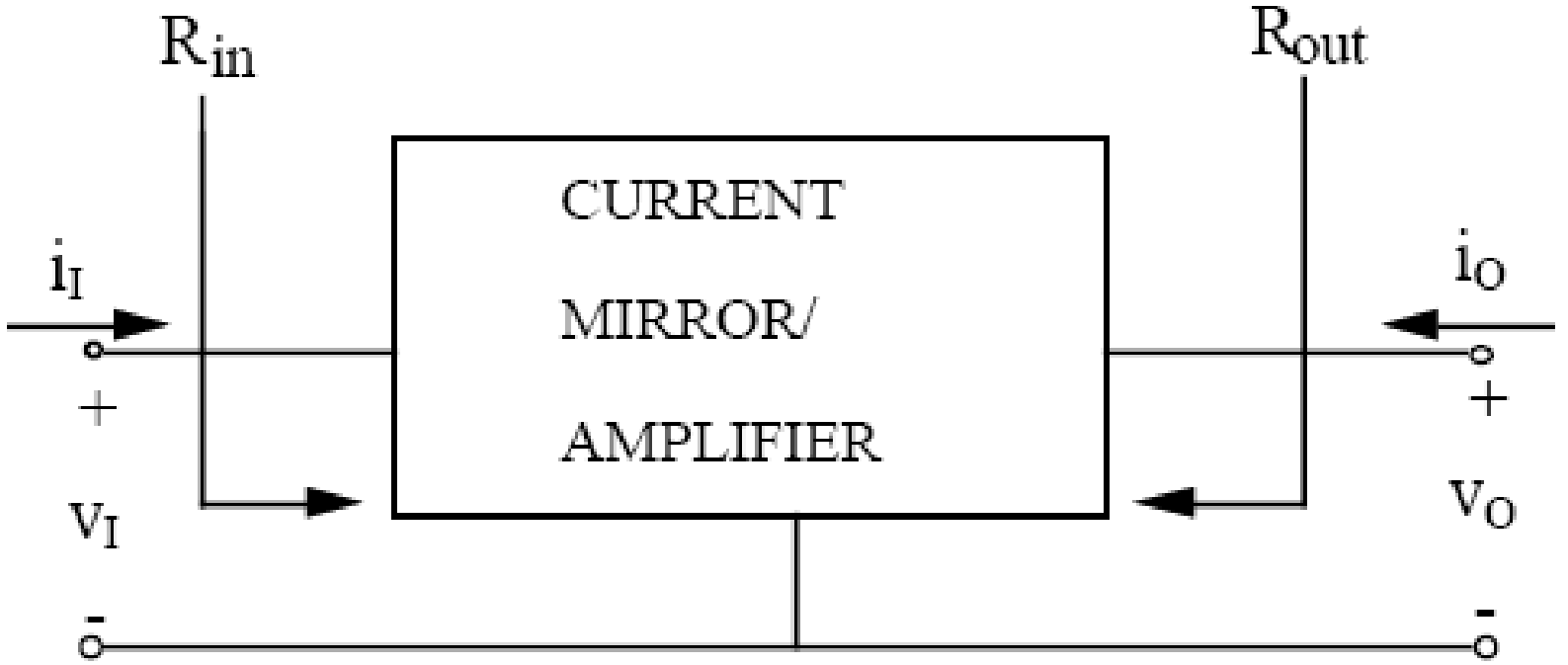
Major sub circuits in analog IC design

- MOS Switch
- Current mirrors
- Single stage amplifiers
  - Common source amplifier
  - Common drain amplifier
  - Common gate amplifier
- DA
- Opamp

Current sources / sinks



Current Mirrors/Current Amplifiers

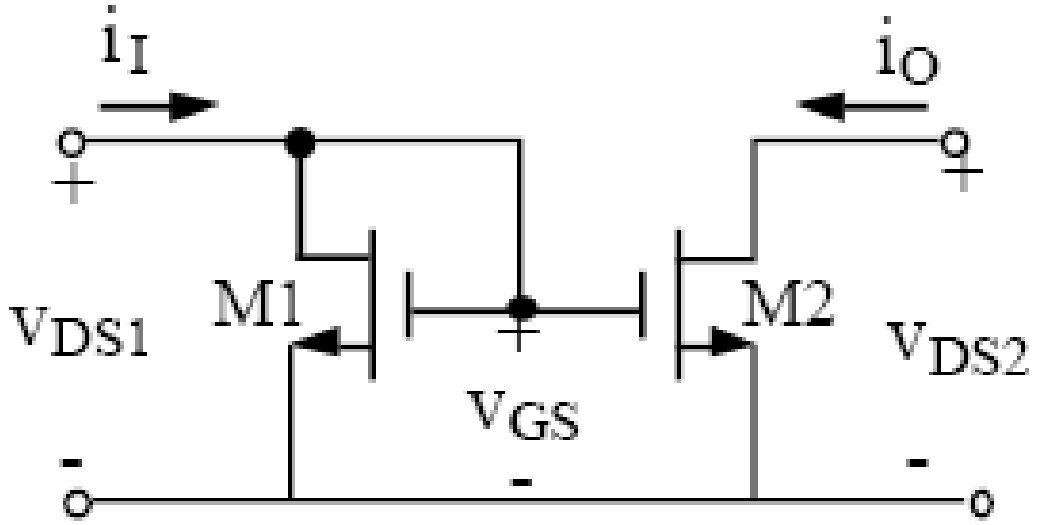


Ideally,

$$i_O = A_I\, i_I$$

$$R_{in} \approx 0 \qquad R_{out} \approx \infty$$

Simple Current Mirrors



Assuming square law model:

$$\frac{i_O}{i_I} = \left(\frac{W_2L_1}{W_1L_2}\right)\left(\frac{v_{GS} - V_{T2}}{v_{GS} - V_{T1}}\right)^2 \left(\frac{1+\lambda v_{DS2}}{1+\lambda v_{DS1}}\right)\left(\frac{\mu_{o2}C_{ox2}}{\mu_{o1}C_{ox1}}\right)$$

If μC_{ox} and V_T matched:

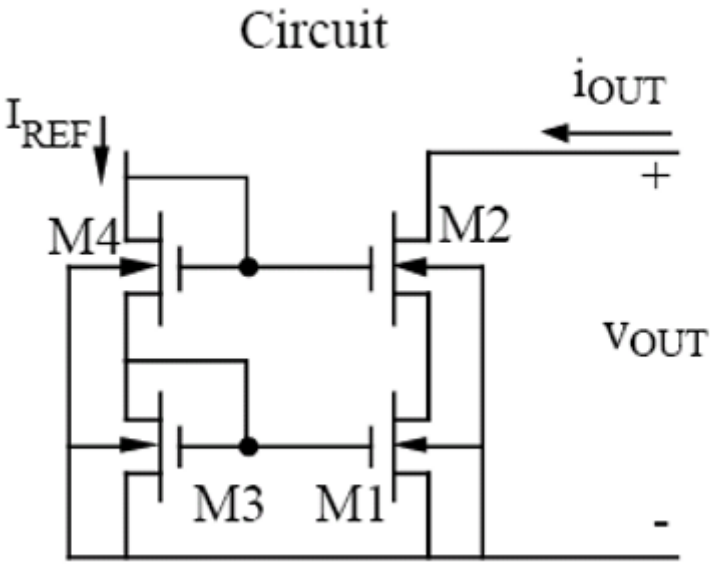
$$\frac{i_O}{i_I} = \left(\frac{W_2L_1}{W_1L_2}\right)\left(\frac{1+\lambda v_{DS2}}{1+\lambda v_{DS1}}\right)$$

If v_{DS} matched:

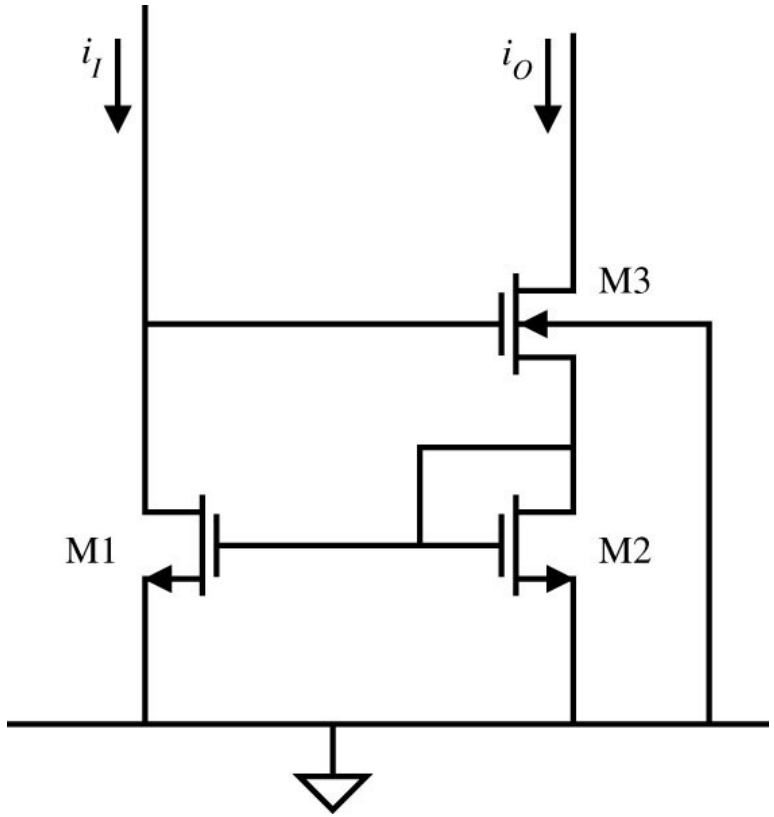
$$\frac{i_O}{i_I} = \frac{W_2L_1}{W_1L_2}$$

Current gain or mirror gain is controlled by geometric ratio, which can be made quite accurate

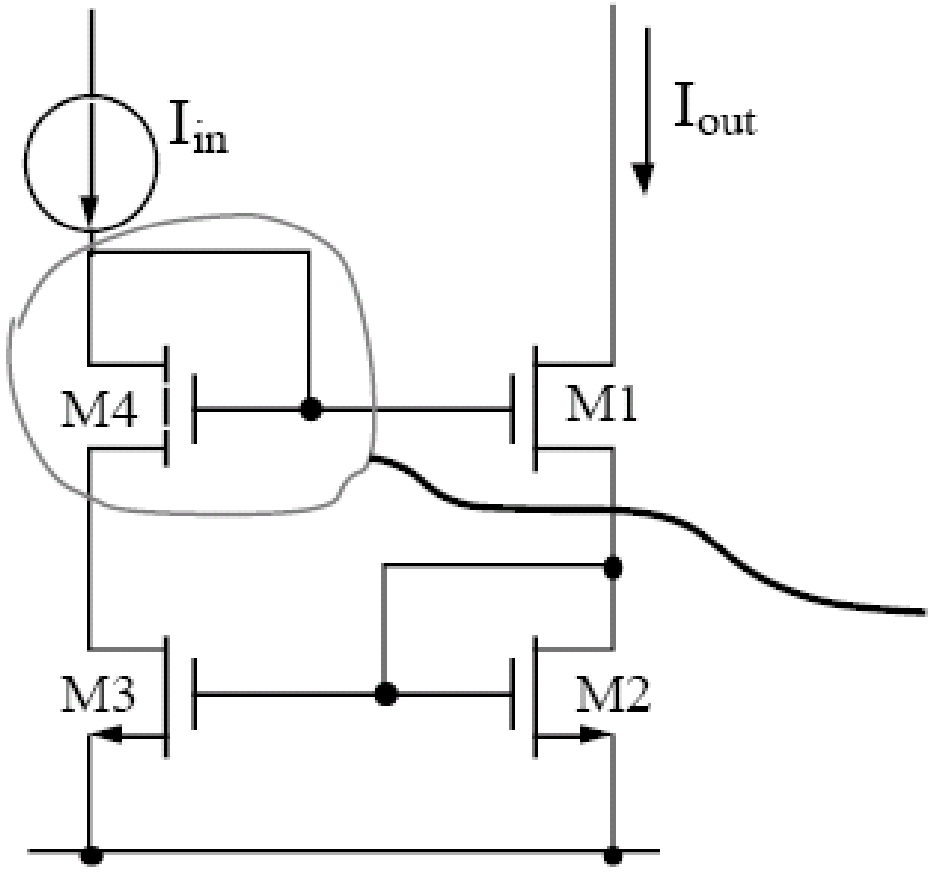
Cascode Current Sink



Wilson Current Mirror



Improved Wilson Current Mirror

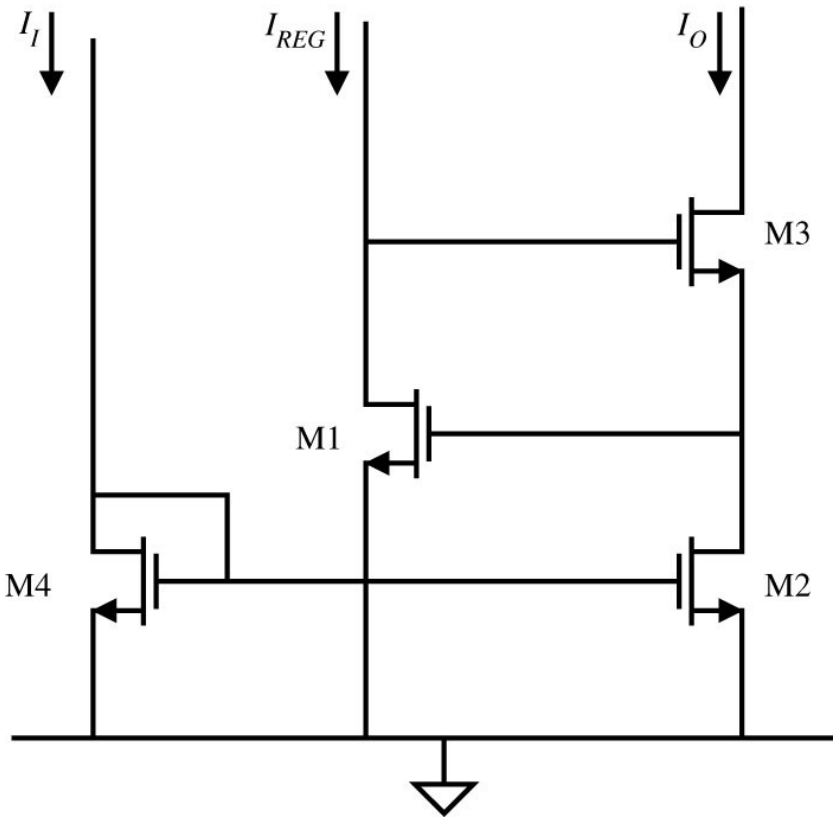


Additional diode-connected transistor equalizes the drain-source voltage drops of transistors M2 and M3

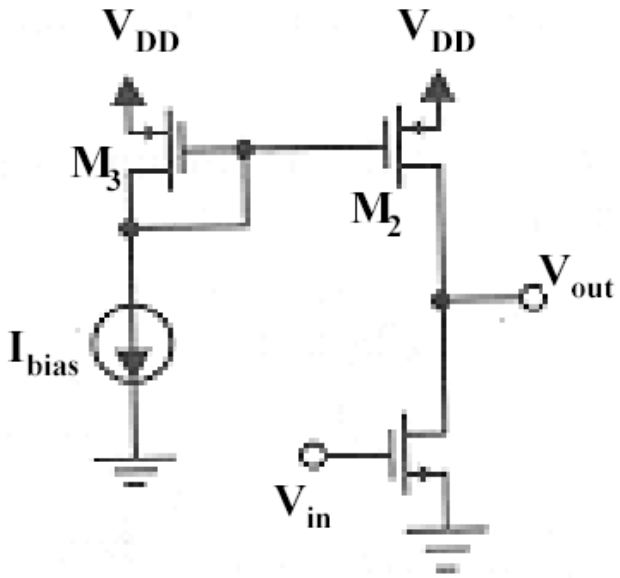
Regulated Cascode Current Mirror

Same as the regulated cascoded current sink

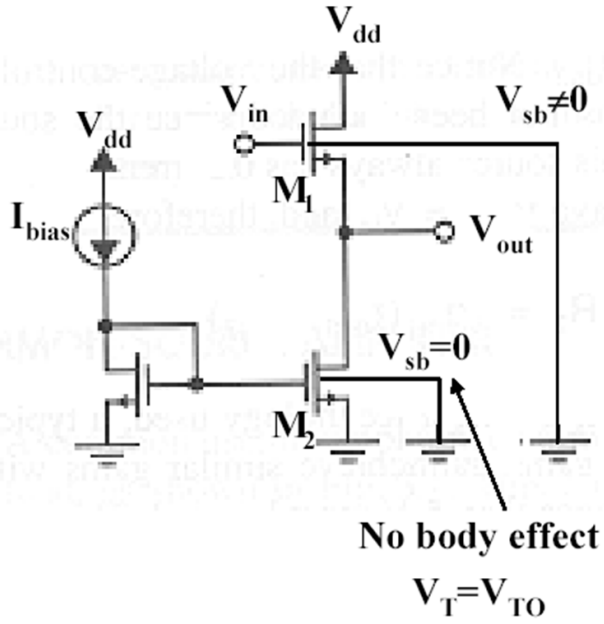
V_{DS2} is very stable with respect to v_o, but not insensitive to I_{reg} change, not necessarily better matching



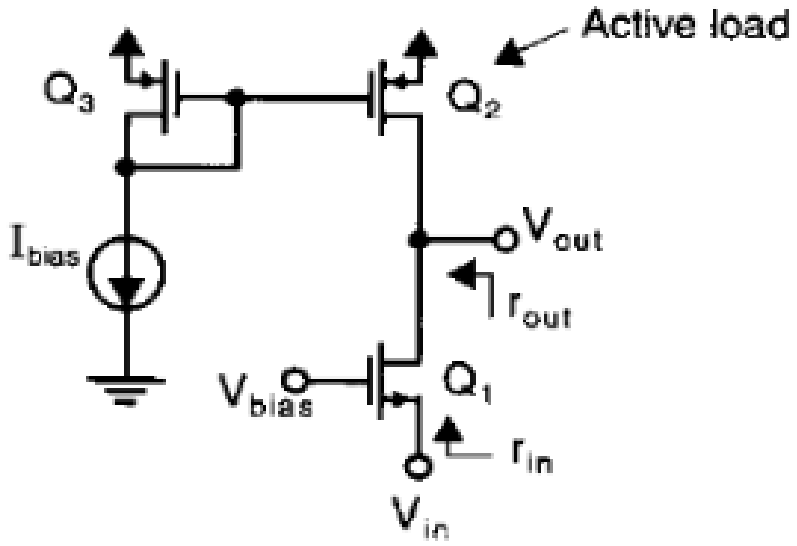
Singal stage amplifier



Common Source Amplifier



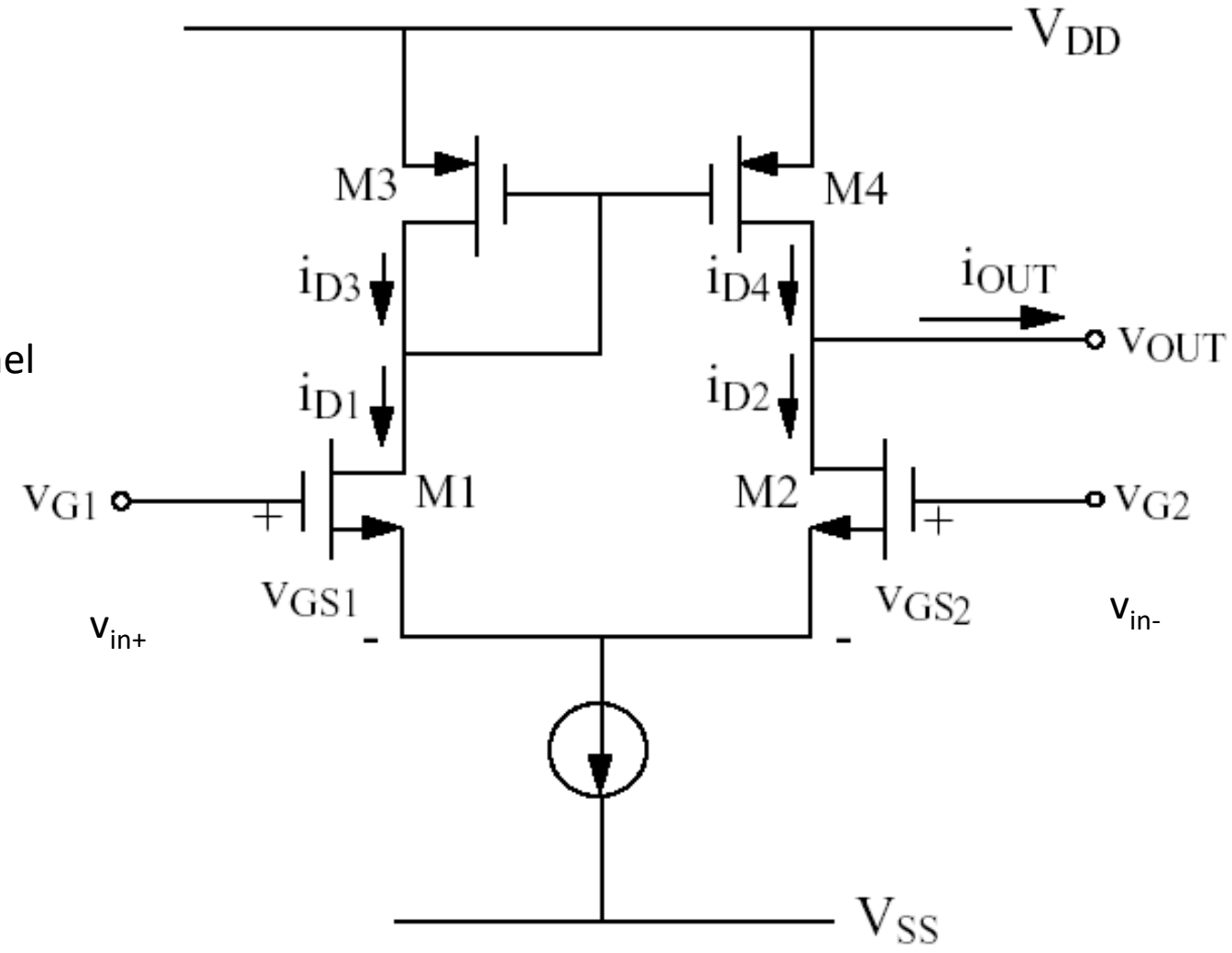
Common Drain Voltage Follower



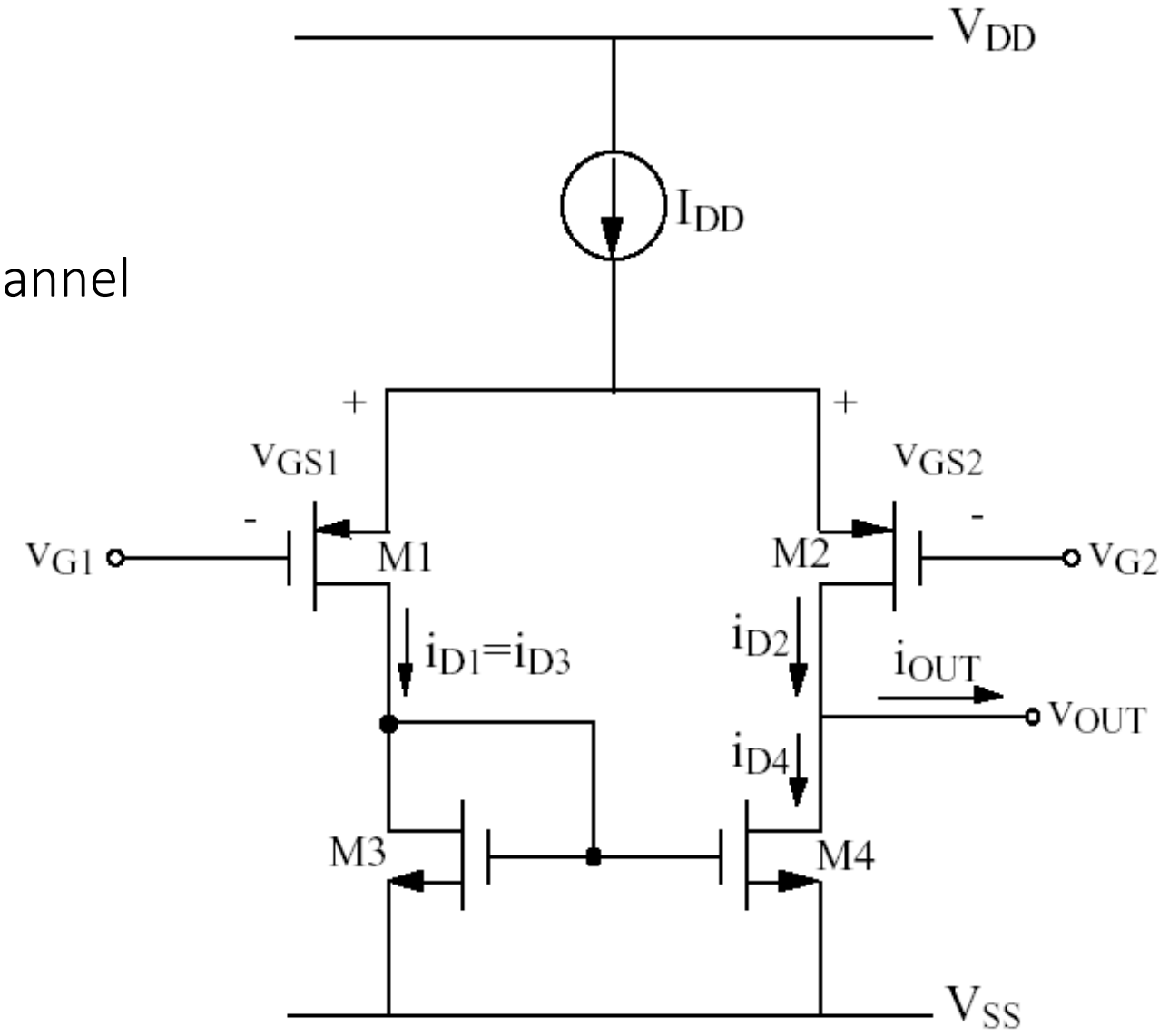
Common Gate Impedance Matching

Differential Input, single-ended output single stage Amplifier

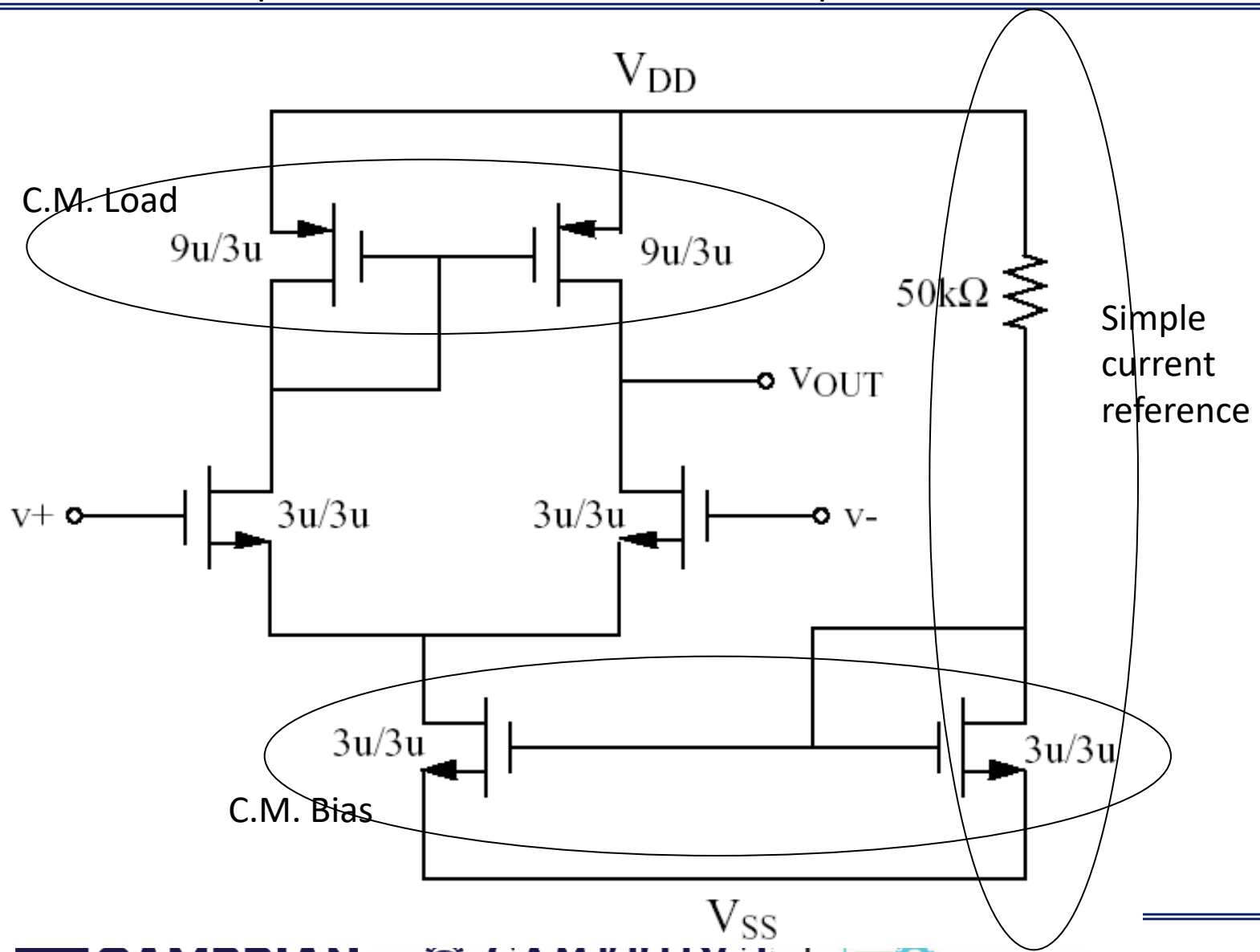
N-Channel



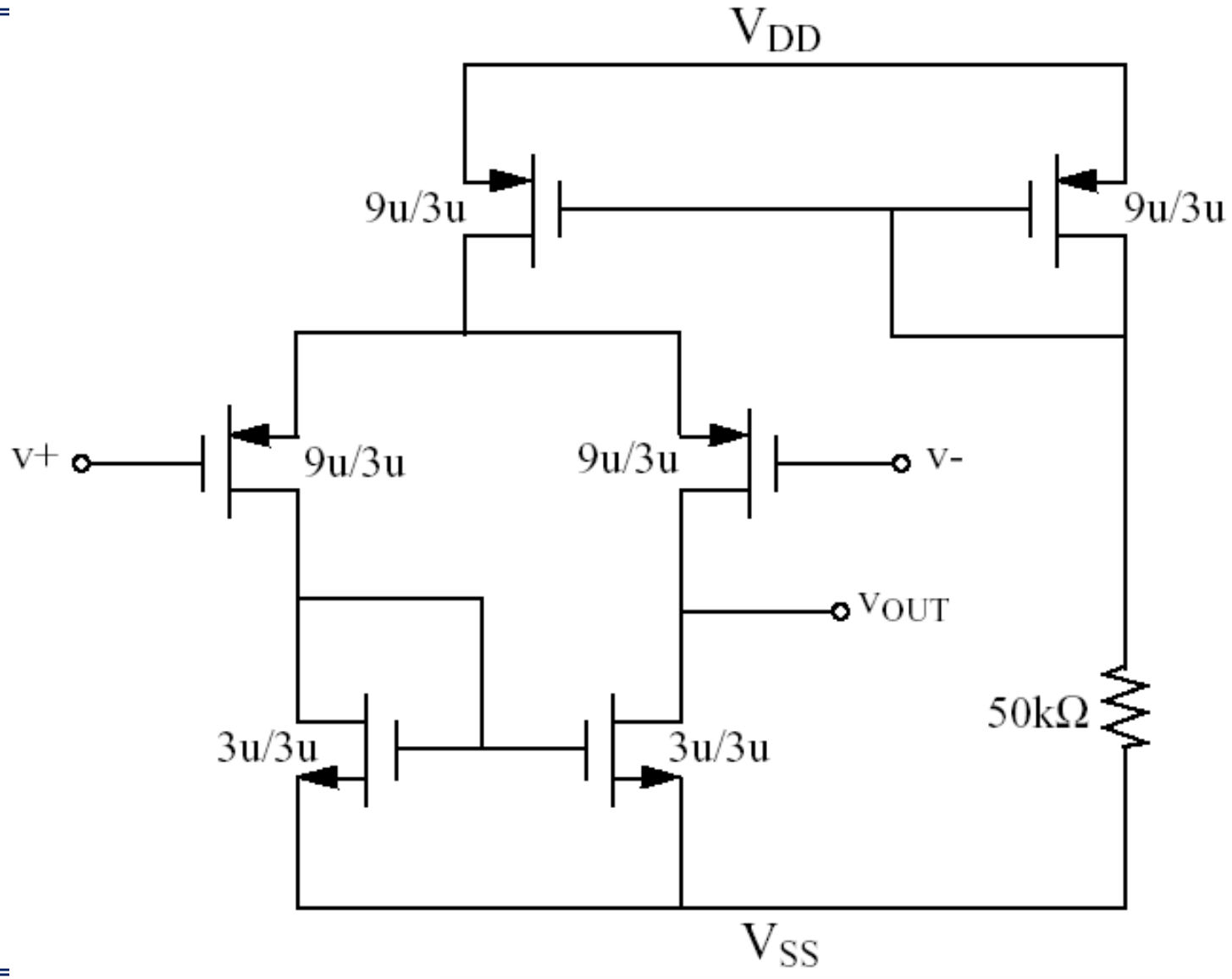
P-channel


VDD
IDD
+
VGS1
-
VG1
M1
+
VGS2
-
M2
VG2
iD1=iD3
iD2
iOUT
VOUT
iD4
M3
M4
VSS

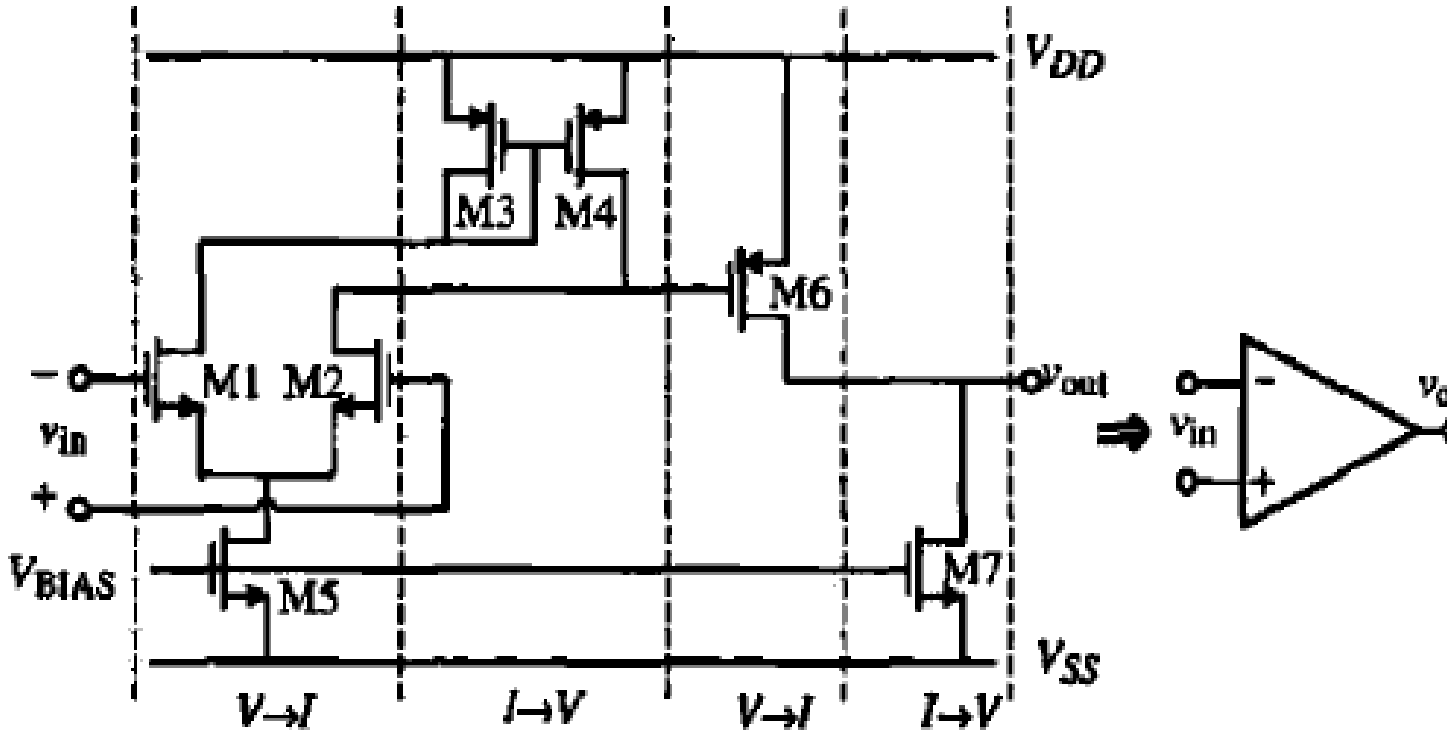
N-Channel Input Pair Differential Amplifier



P-Channel Input Pair Differential Amplifier



Two Stage Op-amp - Design



Case Study: Operational Amplifier Design

Operational Amplifier Design

- Design an two stage operational amplifier with the following specifications:

Av	>5000v/v
Power supply	Vdd = |Vss| = 2.5V
Gain Bandwidth (GB)	5MHz
Load capacitance (CL)	10pf
Slew Rate (SR)	>10v/µs
Vout range	± 2V
ICMR	-1 to 2V
P_{diss}	≤ 2mW

Solution:

Technology Parameters:

Vth= 0.7 v
K′p= 50 µA/V²
K′n= 110 µA/V²

1. To find the desired compensation capacitor from the load capacitor for a 60 phase Margin

Cc > 0.22 CL => Cc > 0.22 * 10 pf
Cc > 2.2pf
Cc =3pf

2. Finding of bias current from the slew rate and compensation capacitor

$$I_5 = SR \cdot Cc = 10\ \mu A * 3pf = 30\ \mu A$$

3. Calculating S3 transistor sizing from the ICMR specifications

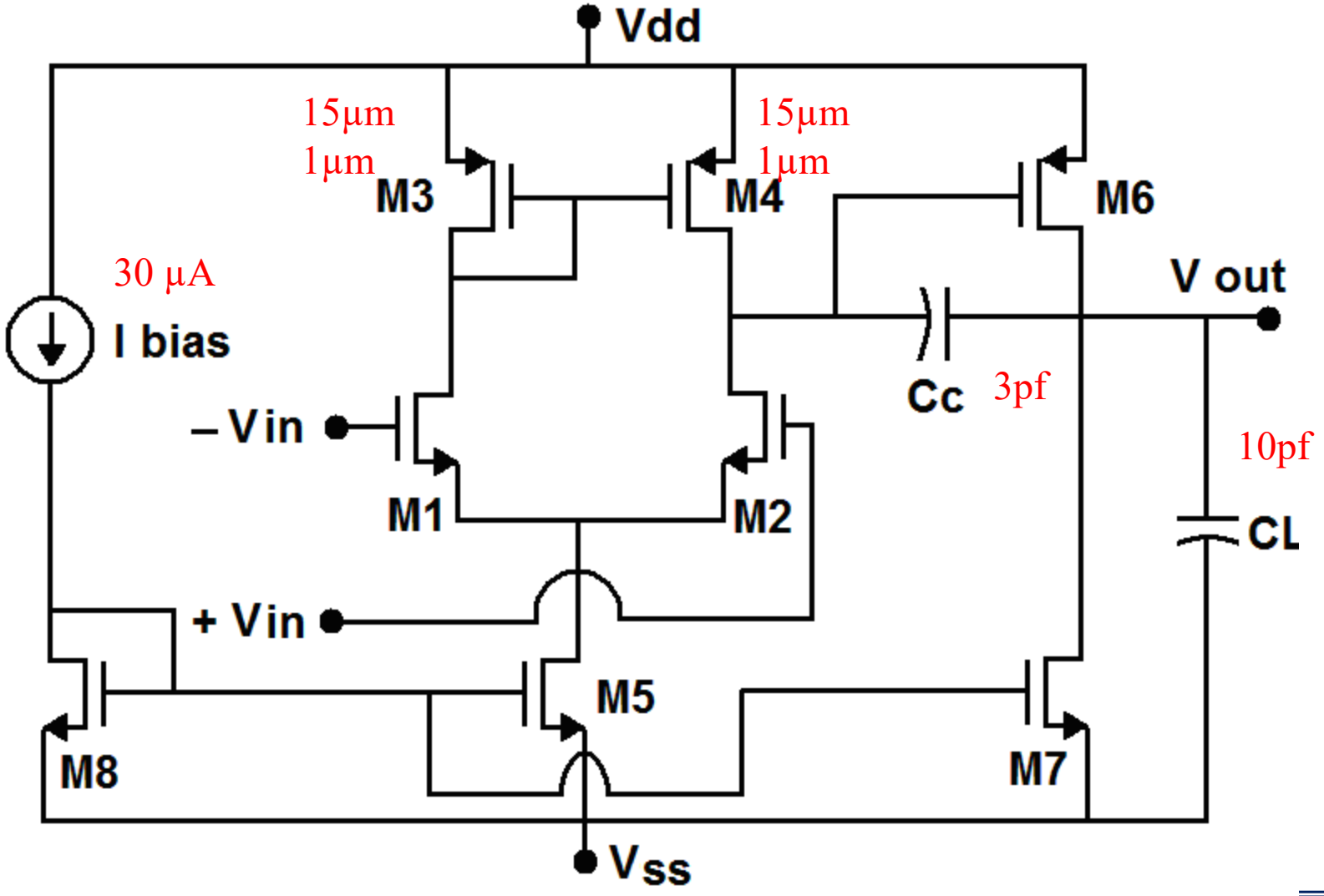
$$S_3 = (W/L)_3 = I5/\ K'3[VDD - Vin\ (max) - |VTO3|(max) + VTL(min)]^2$$

$$= 30x10^{-6}/50x10^{-6}\ [2.5 - 2 - 0.85 + 0.55]^2$$
$$= 15$$

➔ $S_3 = S_4 = 15$

➔ $(W/L)3 = (W/L)_4 = 15$

Operational Amplifier Design cont…



4. Finding Transconductance of the S_1 transistor from the gain bandwidth specification

$$gm_1 = GB.\ Cc = 5X10^6 * 3x10^{12} * 2\pi = 94.25\ \mu s$$

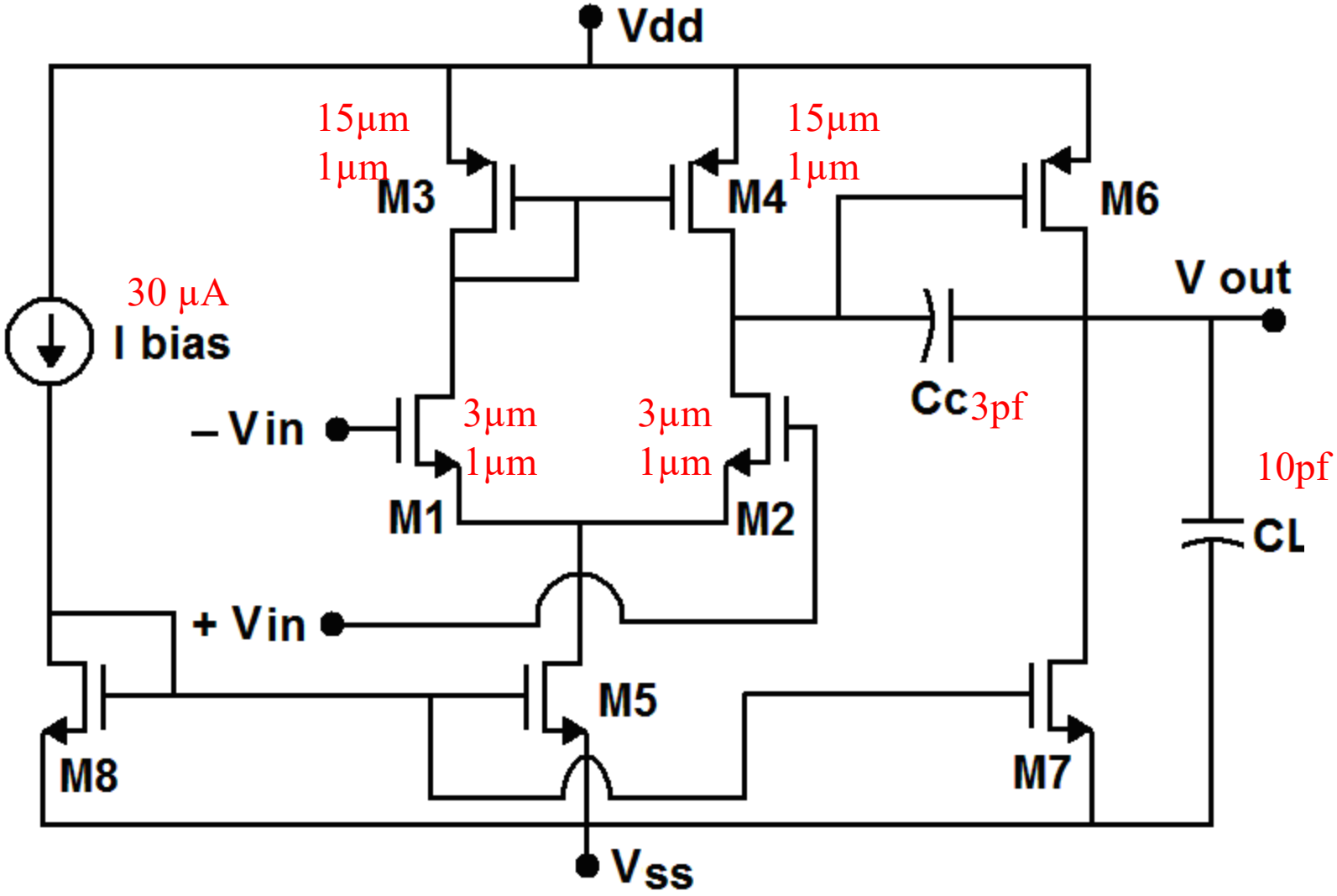
5. Calculating S_1 transistor sizing from the Transconductance

$$S_1 = S_2 = (W/L)_1 = (W/L)_2 = gm_1^2 / 2\ K'_n\ I_1$$
$$= (94.25x10^{-6})^2/2*110x10^{-6}*15X10^{-6}$$
$$= 2.79$$

➔ $S_1 = S_2 = 3$

➔ $(W/L)_1 = (W/L)_2 = 3$

Operational Amplifier Design cont…



6. Calculating VDS of transistor S_5 from ICMR specifications

$$VDS_5 = Vin(min) - V_{ss} - (I_5/\beta_1)^{½} - V_{T(max)}$$
$$= (-1) - (-2.5) - (30x10^{-6}/110x10^{-6}*3)^{½} - 0.85$$
$$= 0.35\ V$$

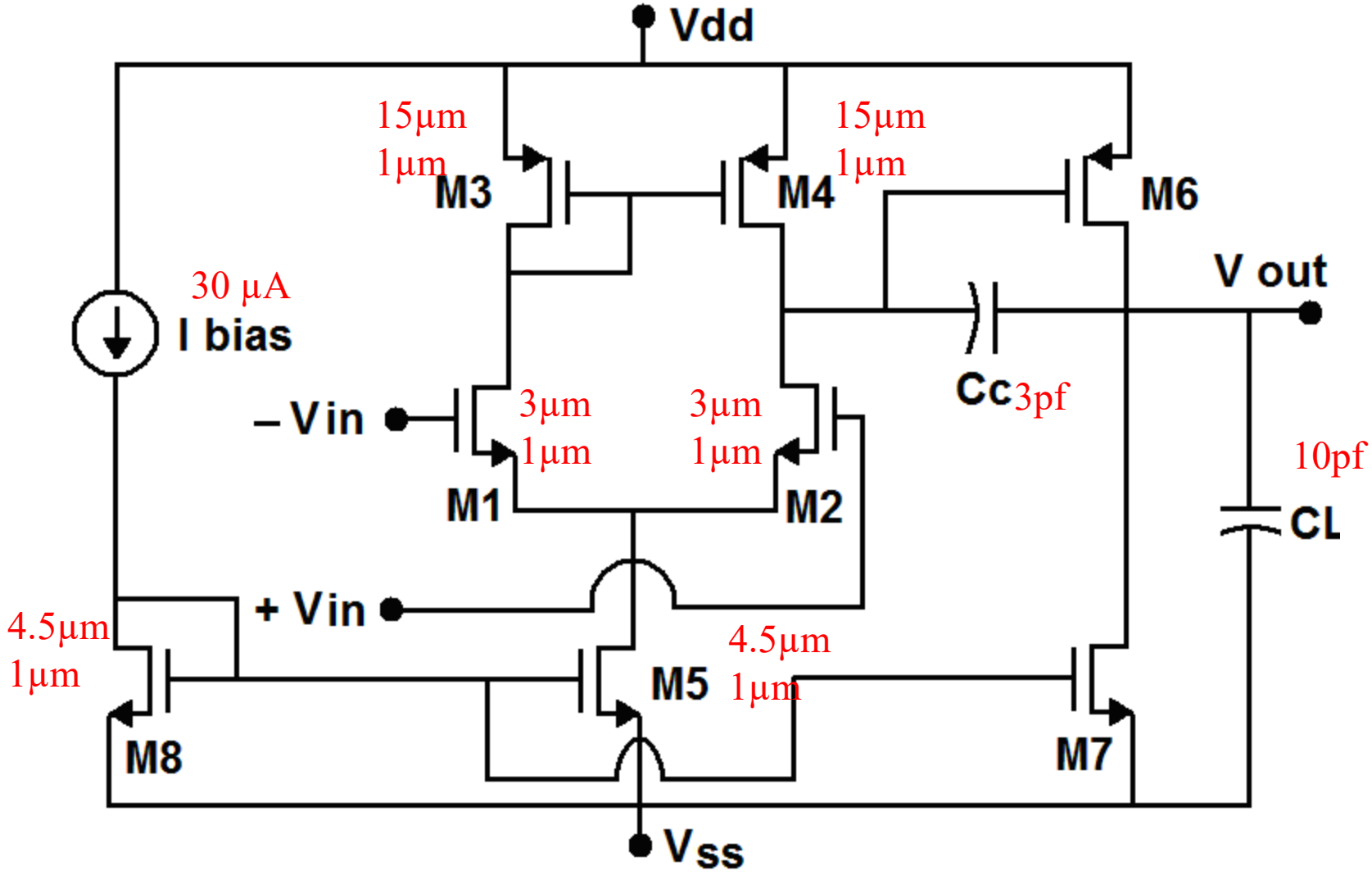
7. $$(W/L)_5 = 2\ I_5/\ K'_5(V_{DS5})^2$$
$$= 2*30x10^{-6}/110x10^{-6}*(0.35)^2$$
$$= 4.49$$

➔ $(W/L)_5 = 4.5$

8. Finding tarnsconductance of S_6 from gm_1

$$gm_6 \geq 10\ gm_1 \Rightarrow gm_6 \geq 10 * 94.25\ \mu$$
$$gm_6 \geq 942.5\mu$$

Operational Amplifier Design cont…



9. Calculating S_6 transistor sizing from the Transconductance of gm_6 and gm_4

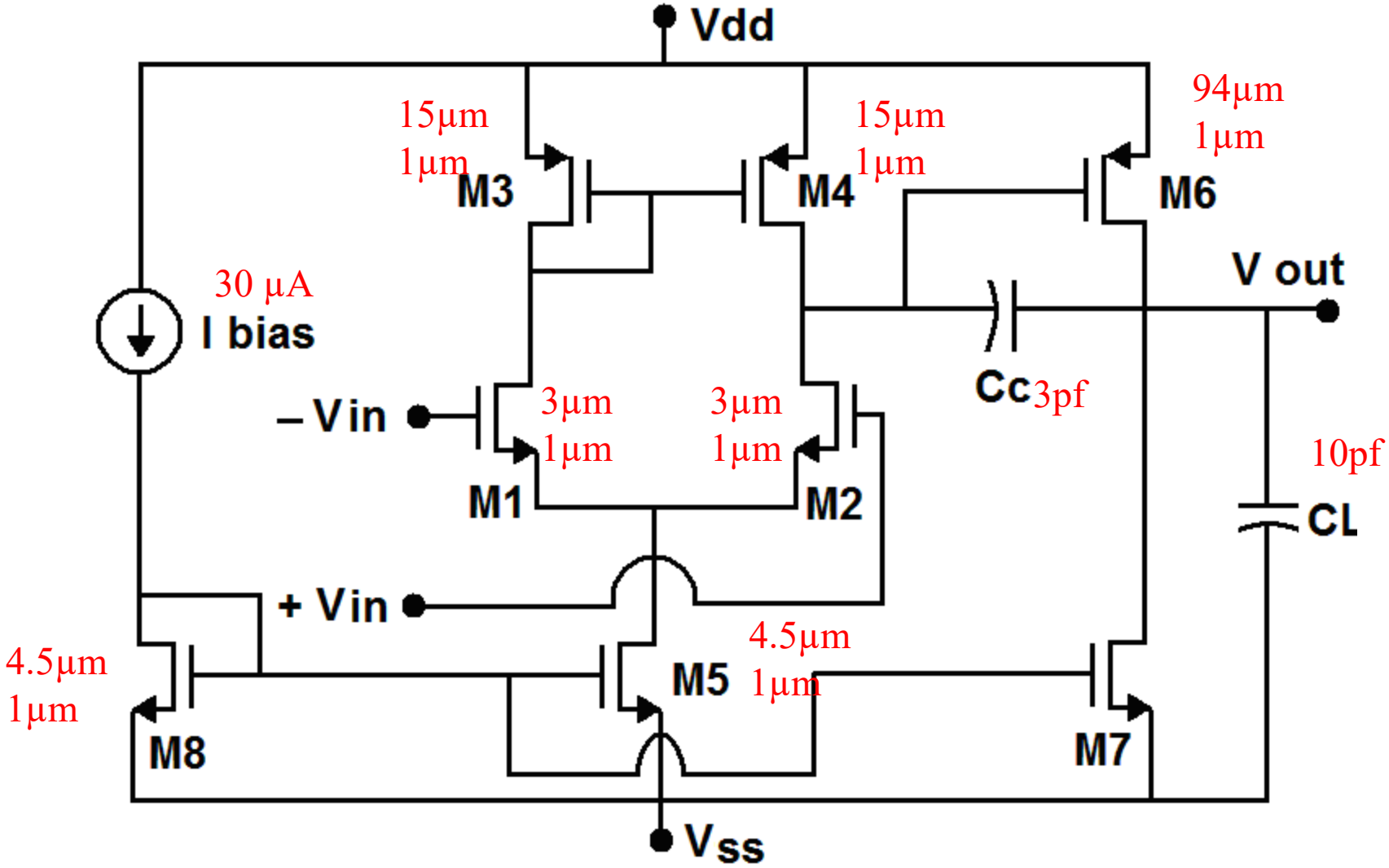
$$(W/L)_6 = (W/L)_4 * gm_6/gm_4$$
$$= 15* 942.5\mu/150\mu$$
$$= 94$$

➔ $(W/L)6= 94$

10. Calculating I_6

$$I6 = (gm_6)^2/2\ K'6(W/L)_6$$
$$= (942.5x10^{-6})^2/2*50x10^{-6}*94 = 94.5uA$$
$$I6 = 95\mu A$$

Operational Amplifier Design cont...



11. Calculating of S_7 from the S_5 ,I_6 and I_5

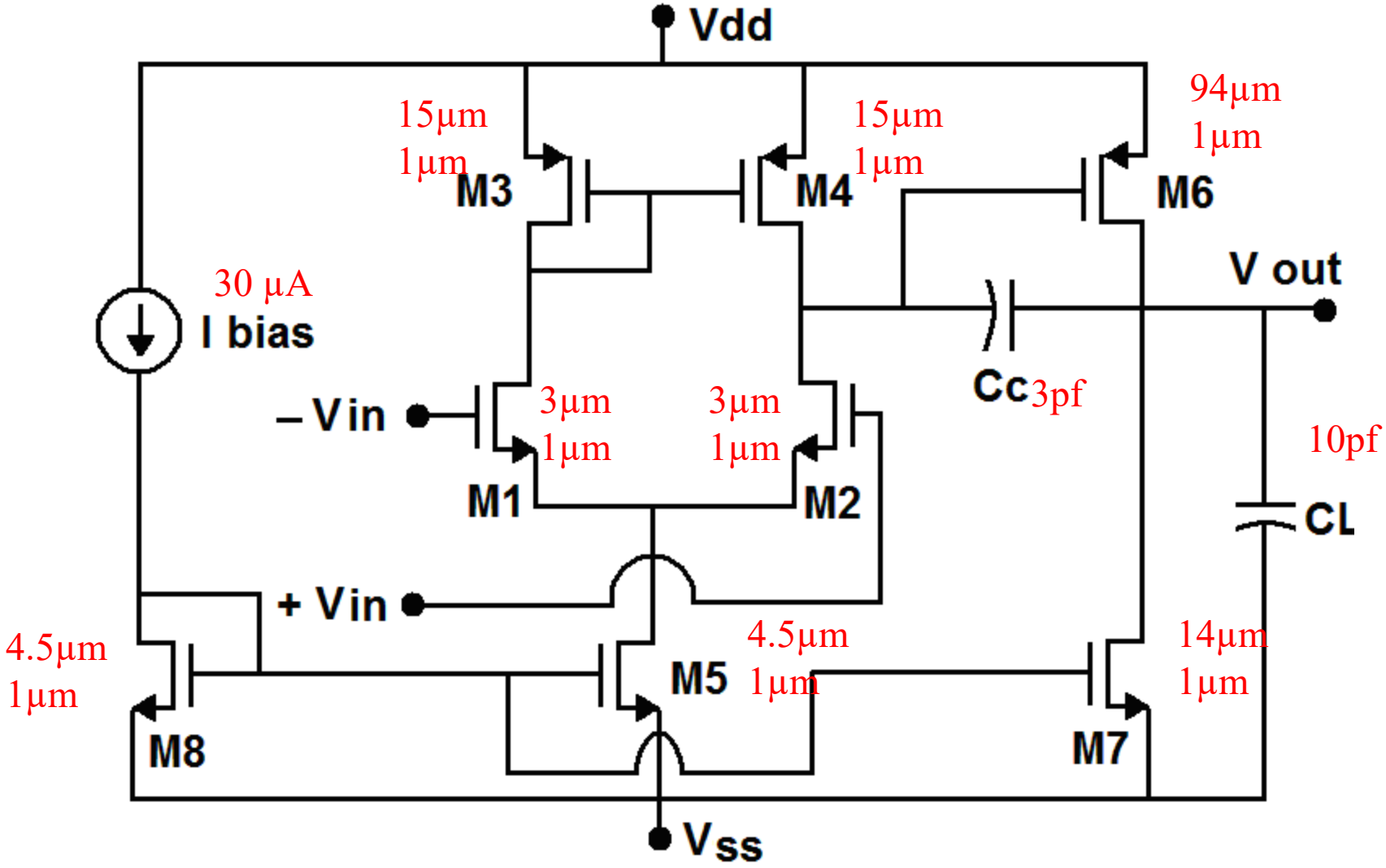
$$(W/L)_7 = (W/L)_5 * I_6 / I_5$$
$$= 4.5 * 95x10^{-6}/30x10^{-6}$$
$$= 14.25$$

➔ $(W/L)7 = 14$

12.

$$Vmin(out) = V_{DS7(sat)} = \sqrt{(2.I_6/ K'(W/L)_7)}$$
$$= \sqrt{2*95x10^{-6}/110x10^{-6}*14}$$
$$= 0.3514V$$

Operational Amplifier Design cont...



13. Calculating the power dissipation of opamp

$$P_{diss} = (I_5 + I_6) *(VDD+|VSS|)$$
$$= (30x10^{-6}+95x10^{-6})*(2.5+2.5)$$
$$P_{diss} = 0.625mW$$

14. Verifying the gain of the two stage operational amplifier

$$A_V = (2.gm_2.gm_6)/I_5 *I_6 *(\lambda_2+ \lambda_3) (\lambda_6+ \lambda_7)$$
$$= 2*94.25\mu*942.5\mu/95\mu*30\mu* (0.04+0.05) (0.05+0.04)$$
$$A_v = 7696v/v$$

Spice Deck for Open Loop Transfer Characteristic

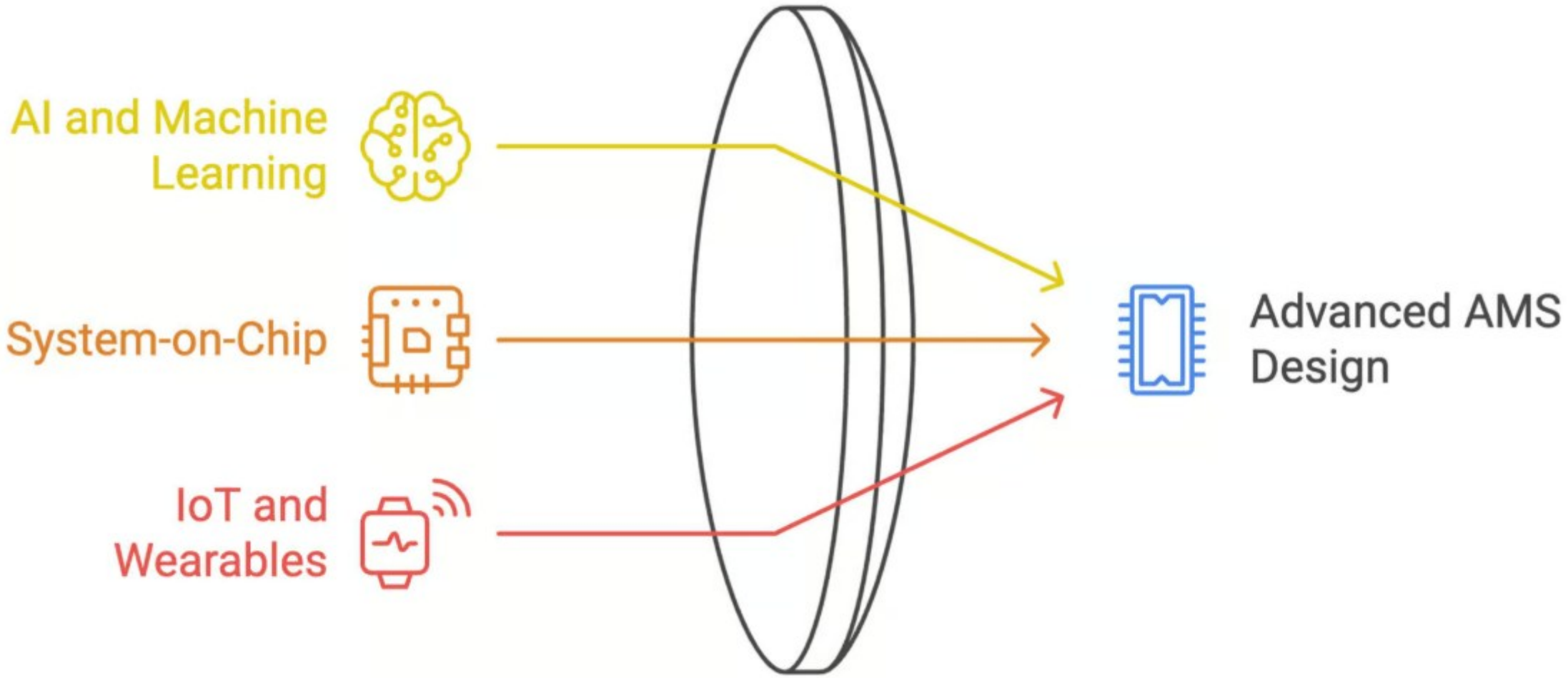
```
CMOS Two Stage Opamp
vin+ 1 0 dc 0 ac 1
vin- 2 0 dc 0
vdd 4 0 dc 2.5
vss 0 5 dc 2.5
.subckt opamp 1 2 6 8 9
m1 4 2 3 3 CMOSN l=1u w=3u
m2 5 1 3 3 CMOSN i=1u w=3u
m3 4 4 8 8 CMOSP l=1u w=15u
m4 5 4 8 8 CMOSP l=1u w=15u
m5 3 7 9 9 CMOSN l=1u w=4.5u
m8 7 7 9 9 CMOSN l=1u w=4.5u
m6 6 5 8 8 CMOSP l=1u w=94u
m7 6 7 9 9 CMOSN l=1u w=14u
ibias 8 7 dc=30u
cc 5 6 3p
.ends
```

```
x1 1 2 3 4 5 opamp
cload 3 0 10p
.op
.dc vin+ -0.005 0.005 100u
.plot dc v(3)
.ac dec 10 1 100meg
.plot ac vdb(3) vp(3) vm(3)
.end
```

Comparison of the Simulation with Design Spec

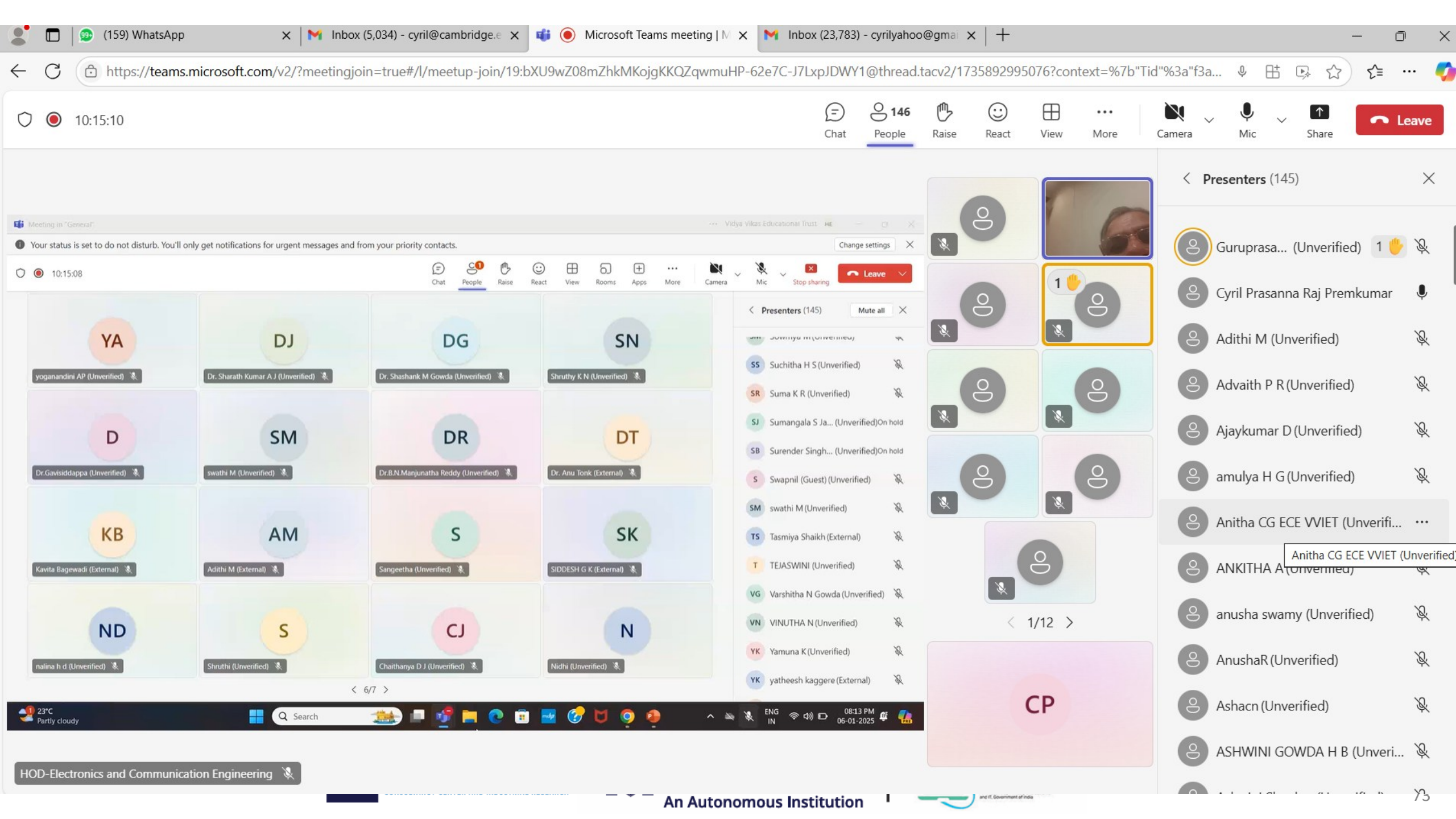
Specification	Design	Simulation
Open loop Gain	> 5000	10,000
GB (MHz)	5 MHz	5 MHz
ICMR (Volts)	-1 to 2 V	+2.4V, -1.2V
Slew Rate (V/us)	>10(V/us)	+10, -7 (V/us)
Pdiss (mw)	<2 mW	0.625 mW
Vout range (V)	±2 V	+2.3V, -2.2V
$PSRR^+$ (dB)	----	87
$PSRR^-$ (dB)	----	116
Phase Margin (Degrees)	60°	65°
Output Resistance (KΩ)	----	122.5 KΩ

Future Trends in AMS Design



Essential Skills for Analog Design Engineers

- **Technical Skills**
- Analog Circuit Design
- Schematic Capture and PCB Design
- Signal and Power Integrity
- Simulation and Analysis


(159) WhatsApp
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Microsoft Teams meeting | M
Inbox (23,783) - cyrilyahoo@gma
https://teams.microsoft.com/v2/?meetingjoin=true#/l/meetup-join/19:bXU9wZ08mZhkMKojgKKQZqwmuHP-62e7C-J7LxpJDWY1@thread.tacv2/1735892995076?context=%7b"Tid"%3a"f3a...
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People
146
Raise
React
View
More
Camera
Mic
Share
Leave
Your status is set to do not disturb. You'll only get notifications for urgent messages and from your priority contacts.
Change settings
10:15:08
Rooms
Apps
Stop sharing
Presenters (145)
Mute all
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yoganandini AP (Unverified)
DJ
Dr. Sharath Kumar A J (Unverified)
DG
Dr. Shashank M Gowda (Unverified)
SN
Shruthy K N (Unverified)
D
Dr.Gavisiddappa (Unverified)
SM
swathi M (Unverified)
DR
Dr.B.N.Manjunatha Reddy (Unverified)
DT
Dr. Anu Tonk (External)
KB
Kavita Bagewadi (External)
AM
Adithi M (External)
S
Sangeetha (Unverified)
SK
SIDDESH G K (External)
ND
nalina h d (Unverified)
S
Shruthi (Unverified)
CJ
Chaithanya D J (Unverified)
N
Nidhi (Unverified)
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SS Suchitha H S (Unverified)
SR Suma K R (Unverified)
SJ Sumangala S Ja... (Unverified)On hold
SB Surender Singh... (Unverified)On hold
S Swapnil (Guest) (Unverified)
SM swathi M (Unverified)
TS Tasmiya Shaikh (External)
T TEJASWINI (Unverified)
VG Varshitha N Gowda (Unverified)
VN VINUTHA N (Unverified)
YK Yamuna K (Unverified)
YK yatheesh kaggere (External)
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CP
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Guruprasa... (Unverified)
Cyril Prasanna Raj Premkumar
Adithi M (Unverified)
Advaith P R (Unverified)
Ajaykumar D (Unverified)
amulya H G (Unverified)
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ANKITHA A (Unverified)
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